



**VIDYA VIKAS INSTITUTE OF
ENGINEERING AND TECHNOLOGY,
MYSORE**

LIBRARY AND INFORMATION CENTRE

VTU Previous Question Papers

BE – CV

III to VIII Semester

**Dec 2024 /Jan -2025
2021, 2022 Scheme**

VVIET

Bannur road, Alanahalli, Mysore-570028

INDEX

Sl.No.	Sub-Code	Subject Title	Exam Date
1	21CV384	Infrastructure Finance	Dec.2024/Jan.2025
2	21CV43	Public Health Engineering	Dec.2024/Jan.2025
3	21CV51	Hydrology and Water Resource Engineering	Dec.2024/Jan.2025
4	21CV52	Transportation Engineering	Dec.2024/Jan.2025
5	21CV53	Design of RC Structural Elements	Dec.2024/Jan.2025
6	21CV54	Geotechnical Engineering	Dec.2024/Jan.2025
7	21CIV57	Environmental Studies	Dec.2024/Jan.2025
8	21CV646	Alternative Building Materials	Dec.2024/Jan.2025
9	21CV71	Quantity Survey and Contract Management	Dec.2024/Jan.2025
10	21CV72	Construction Technology for Substructures and Super	Dec.2024/Jan.2025
11	21CV731	Advanced Design of RCC and Steel Structures	Dec.2024/Jan.2025
12	21CV742	Ground Improvement Technique	Dec.2024/Jan.2025
13	BCV301	Strength of Materials	Dec.2024/Jan.2025
14	BCV302	Engineering Survey	Dec.2024/Jan.2025
15	BCV303	Engineering Geology	Dec.2024/Jan.2025
16	BCV304	Water Supply and Wastewater Engineering	Dec.2024/Jan.2025
17	BCV306C	Sustainable Design Concept for Building Services	Dec.2024/Jan.2025
18	BCV401	Analysis of Structures	Dec.2024/Jan.2025
19	BCV402	Fluid Mechanics and Hydraulics	Dec.2024/Jan.2025
20	BCV403	Transportation Engineering	Dec.2024/Jan.2025
21	BCV456C	Electronic Waste Management - Issues and Challenges	Dec.2024/Jan.2025
22	BBOK407	Biology for Engineers	Dec.2024/Jan.2025
23	BUHK408	Universal Human Values	Dec.2024/Jan.2025
24	BCV501	Construction Management and Entrepreneurship	Dec.2024/Jan.2025
25	BCV502	Geotechnical Engineering	Dec.2024/Jan.2025
26	BCV503	Concrete Technology	Dec.2024/Jan.2025
27	BESK508	Environmental Studies	Dec.2024/Jan.2025
28	BCV515C	Solid Waste Management	Dec.2024/Jan.2025
29	BRMK557	Research Methodology and IPR	Dec.2024/Jan.2025

USN								
-----	--	--	--	--	--	--	--	--

Question Paper Version : A

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Infrastructure Finance

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

- Which of the following is not in the infrastructure sector?
 - Power generation
 - Construction of roads
 - Food production
 - Expansion of ports
- Which of the following is not a "Public good"?
 - Electricity
 - National defense
 - Light house
 - Public parks
- Which of the following is not an economic activity?
 - Voluntary social service
 - Forming
 - Transportation
 - Service
- _____ is an example of social infrastructure
 - Transport
 - Communication
 - Energy
 - Education
- Which of the following statements is true about social infrastructure?
 - Electricity is an essential component of social infrastructure
 - Roads and highways are an essential component of social infrastructure.
 - Internet is an essential component of social infrastructure.
 - Housing is an essential component of social infrastructure.
- The _____ continues to be largest employer in India
 - Quaternary sector
 - Tertiary sector
 - Secondary sector
 - Primary sector

7. Workers in _____ sector are under employed
 - a) IT
 - b) Agriculture
 - c) Manufacturing
 - d) None of these
8. _____ Shows how big the economy of a country is.
 - a) GSDP
 - b) GDP
 - c) PPP
 - d) None of these
9. The unorganized sector is characterized by small and scattered units which are largely outside the control of the _____.
 - a) Employers
 - b) Private individuals
 - c) Government
 - d) Employers
10. Challenges faced by infrastructure projects
 - a) Land acquisition
 - b) Access to financing
 - c) Political and regulatory risk
 - d) All of these
11. Types of infrastructure sector
 - a) Roads, Rails and Aviation
 - b) Multimodel logistics
 - c) Shipping
 - d) All of these
12. Cash out flows include
 - a) Debts
 - b) Liabilities
 - c) Operating cost
 - d) All of these
13. Capital flows will be effectively working under
 - a) Government sector
 - b) Private sector
 - c) Office sector
 - d) Semi official sector
14. What does working capital management manage?
 - a) Long term assets
 - b) Long term liabilities
 - c) Short term liabilities and assets
 - d) Only short term liabilities
15. Which one is/are financial assets?
 - a) Bonds
 - b) Machines
 - c) Stocks
 - d) Both a and c
16. Which of the following is a secondary source of energy?
 - a) Hydrocarbons
 - b) Oil
 - c) Both a and b are correct
 - d) Both a and b are incorrect
17. Which of these is not a part of capital structure?
 - a) Equity shares
 - b) Debentures
 - c) Short term borrowings
 - d) Bonds
18. Which of the above factors help to determine capital structure of a firm?
 - a) Government policies
 - b) Degree of control
 - c) Cost of capital
 - d) All of these
19. The process of financing the assets of a business is known as
 - a) Asset structure
 - b) Owners structure
 - c) Financial structure
 - d) Capital structure

20. Capital structure _____ financial structure
a) is a part of
b) is not a part of
c) is the same as
d) is different from
21. The increase in the stock of the capital is also called
a) Capital Profit
b) Capital formation
c) Capital loss
d) None of these
22. Which one of the following options is not considered as flow?
a) Depreciation
b) Investment
c) Income
d) Capital
23. Gross profit is
a) Cost of goods sold + opening stock
b) Sales – cost of goods sold
c) Sales – purchases
d) Net profit – expenses
24. Investment in human capital is expenditure on
a) Education
b) Training
c) Medical care
d) All of these
25. Formula source of credit does not include
a) Banks
b) Co operatives
c) Employers
d) Regional rural banks
26. Which of the following statements are true regarding the organized sector?
a) Workers in the organized sector enjoy the security of employment.
b) They are expected to work only a fixed number of hours.
c) If they work more, they have to be paid overtime by the employer.
d) All of these
27. How does infrastructure contribute to economic development of a country?
a) Increasing the productivity of the factors of production.
b) Improving the quality of life of its people.
c) Both A and B
d) None of these
28. In an organization the share holders wealth is represented by
a) The salary paid to employees
b) The market price of share
c) The book value of firm's assets
d) None of these
29. Agriculture, dairy, fishing, and forestry are examples of
a) Tertiary sector
b) Secondary sector
c) Primary sector
d) None of these
30. The market price of an equity share is determined by
a) The president of a company
b) The board of directors
c) Buyers and sellers of those shares
d) The stock exchange where those shares are getting traded.

43. The capital market in India is controlled by
a) RBI b) NABARD c) SEBI d) IRDA
44. Which of the above factors helps to determine the capital structure of a firm?
a) Government policies b) Degree of control
c) Cost of capital d) All of these
45. _____ is a leading indicator of market behavior
a) Arbitrage b) Index
c) Investment d) None of these
46. The ability to convert an asset rapidly and without influencing its price is referred to as

a) Scalability b) Liquidity
c) Marketability d) Minimal risk
47. There is an increase in demand for good quality infrastructure because of
a) Economic growth b) Modernization
c) Health issues d) All of these
48. Economic infrastructure supports the economy _____
a) Indirectly b) Directly
c) Doesn't effect d) None of these
49. When was NIFTY established?
a) 1992 b) 1998 c) 1996 d) 1994
50. The headquarters of the National Stock exchange is situated in
a) Mumbai b) Kolkata
c) Chennai d) Delhi

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

21CV43

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Public Health Engineering

Time: 3 hrs.

Max. Marks: 100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Any missing data assumed suitably.**

Module-1

- 1 a. Write the needs of protected water supply. (04 Marks)
- b. Briefly explain various types of water demands. (08 Marks)
- c. Briefly explain factors affecting Per Capita Demand. (08 Marks)

OR

- 2 a. Mention the methods of forecasting population and explain any two of them. (08 Marks)
- b. Write the physico-chemical characteristics of water. (06 Marks)
- c. Compute the population of the year 2000 and 2006 for a city where population in the year 1930 was 25,000 and in the year 1970 was 47000 using geometric increase method. (06 Marks)

Module-2

- 3 a. With the help of general flow diagram of water treatment plant, explain the function of each unit. (08 Marks)
- b. Explain the types of sedimentation. (06 Marks)
- c. Explain the methods of Aeration. (06 Marks)

OR

- 4 a. Define coagulation and flocculation and write the types of coagulants commonly used in water treatment. (06 Marks)
- b. A water work has to purify the water for a town whose daily demand is 9×10^6 lit/day. Design the suitable rectangular shape sedimentation tank of the water works filled with mechanical sludge remover. Assume the velocity of flow in the sedimentation tank as 0.22 m/sec and the detention time as 8 hours. (08 Marks)
- c. Write the comparison between a slow sand filter and rapid sand filter. (06 Marks)

Module-3

- 5 a. Explain the methods of disinfection. (08 Marks)
- b. With the help of graph, explain breakpoint chlorination. (06 Marks)
- c. Explain lime soda process and zeolite process for hardness removal. (06 Marks)

OR

- 6 a. Briefly explain methods of sewage disposal. (06 Marks)
- b. Briefly explain types of water carriage system. (08 Marks)
- c. Write the physical, chemical and biological characteristics of waste water. (06 Marks)

Module-4

- 7 a. With the help of general flow diagram of waste water treatment plant, explain the functions of each unit. (08 Marks)
- b. Explain the types of screen used in wastewater treatment. (06 Marks)
- c. Design a bar screen for a peak average flow of 40 MLD with the following data:
Angle of inclination = 45° to horizontal bar size 9 mm $\times$ 50 mm , clear spacing 36 mm,
Desired velocity through the screen = 0.8 m/sec. (06 Marks)

OR

- 8 a. Explain unit operations and processes in waste water treatment. (04 Marks)
- b. With the help of neat sketch, explain circular type of settling tank. (08 Marks)
- c. With the help of flow diagram, explain the working of activated sludge process. (08 Marks)

Module-5

- 9 a. Explain impended growth system and attached growth system in waste water treatment. (04 Marks)
- b. With the help of neat sketch, explain construction and working of trickling filter. (10 Marks)
- c. Explain the working of rotating biological contactors. (06 Marks)

OR

- 10 a. Write the advantages and disadvantages of stabilization ponds. (06 Marks)
- b. With the help of sketch, explain working of sludge drying beds. (08 Marks)
- c. Write the advantages and disadvantages of aerobic digestion. (06 Marks)

* * * * *

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

21CV51

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Hydrology and Water Resource Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain hydrologic cycle (Horton's) with process involved in it. (10 Marks)
b. What are the importance of hydrology with emphasis of global water availability. (10 Marks)

OR

- 2 a. Define rain gauge. Describe with a neat sketch the principle of working of Symon's non-recording gauge. (10 Marks)
b. A 12 – hour storm rainfall with the following depth in cm occurred over a basin :
2.0, 2.5, 7.6, 3.8, 10.6, 5.0, 7.0 10.0, 6.4, 3.8, 1.4 and 1.4.
The surface run – off resulting from the above storm is equivalent to 25.5 cm of depth over the basin. Determine the average infiltration index for the basin (10 Marks)

Module-2

- 3 a. What is runoff? List and explain Factor's affecting it. (10 Marks)
b. Define the following :
i) Basin recharge
ii) Direct run off
iii) Drainage density
iv) Over land flow. (10 Marks)

OR

- 4 a. Define "unit hydrograph" with the help of neat sketch explain various components of a flood hydrograph. (10 Marks)
b. The hourly ordinates of a two-hour unit hydrograph are given below. Derive a 6 hour unit hydrograph for the same catchment :

Time hours	00	01	02	03	04	05	06	07	08	09	10	11	12	13	14	15
Discharge cumecs	0.0	1.0	2.7	5.0	8.0	9.8	9.0	7.5	6.3	5.0	4.0	2.9	2.1	1.3	0.5	0.0

(10 Marks)

Module-3

- 5 a. Discuss the benefits and ill effects of irrigation. (10 Marks)
b. Explain with a neat sketch Bondhar's irrigation. List out advantages and disadvantages. (10 Marks)

OR

- 6 a. Define duty and delta. Derive the relation between them. (08 Marks)
b. What is irrigation efficiency? Define different efficiencies of irrigation water. (06 Marks)
c. What are flow irrigation and lift irrigation. Explain briefly. (06 Marks)

Module-4

- 7 a. What is canal? List its types and explain with a neat sketch its classification based on alignment. (10 Marks)
b. What is meant by design of canal? Bring out the difference between Kennedy's and Lacey's theory. (10 Marks)

OR

- 8 a. With sketch explain different zones of a storage reservoir. (10 Marks)
b. Explain hydrological investigations of reservoir planning. List the points to be considered for selection of a site for a reservoir. (10 Marks)

Module-5

- 9 a. Explain causes of floods and what are the methods adopted to flood alleviation. (control of floods). (10 Marks)
b. Explain how flood damage analysis is estimated. (10 Marks)

OR

- 10 a. Explain the Drought? Explain the causes of drought and measures for water conservation. (10 Marks)
b. Explain the different methods of rainwater harvesting. (10 Marks)

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

21CV52

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Transportation Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the characteristics of Road transport in comparison with other transport. (06 Marks)
- b. What is Jayakar Committee and what are its recommendation? How is it implemented? (06 Marks)
- c. Determine the length of different categories of roads in a state in India using the 3rd road development plan formulae and the given data :
 - i) Total area of the State = 4,30,000 Sq. km.
 - ii) Total number of towns = 550.
 - iii) Overall road Density aimed at = 75km/100 km². (08 Marks)

OR

- 2 a. List the factors affecting the Alignment. Briefly explain the Engineering surveys for Highway alignment. (06 Marks)
- b. List the silent features of: i) KSHIP ii) NHDP. (06 Marks)
- c. Fix the priority for the following system of roads assuming suitable population and productivity units : (08 Marks)

Road Link	Length (km)	Number of villages with population served			Productivity served in Tonnes	
		<1000	1001 to 2000	> 2000	Agriculture	Industrial
A	35	18	14	6	26000	10000
B	28	13	8	5	16000	12000
C	21	12	5	3	10000	14000

Module-2

- 3 a. Explain briefly any two important surface characteristics influencing highway geometric design. (06 Marks)
- b. Define Super Elevation. Derive an expression for super elevation with neat sketch. (08 Marks)
- c. Determine SSD on a : i) Level road ii) Down grade of 1 in 120.
Take $t = 2.5$ secs , $f = 0.35$, $v = 80$ kmph. (06 Marks)

OR

- 4 a. Draw a neat sketch of flexible pavement cross – section and show the component parts. Enumerate the functions of each components of the pavement. (10 Marks)
- b. Explain 'ESWL' and the graphical method of determination of ESWL for dual wheel assembly. (10 Marks)

Module-3

- 5 a. List the soil classification systems and explain briefly the HRB soil classification system. (07 Marks)
b. What are the desirable properties of road aggregates? Mention the laboratory tests conducted to determine the property. (07 Marks)
c. Distinguish between cutback and emulsion and list the tests on Bitumen. (06 Marks)

OR

- 6 a. Explain the procedure involved in the preparation of sub – grade layer along with quality control tests. (10 Marks)
b. Explain step by step procedure for construction of :
i) Water Bound Macadam ii) Bituminous concrete. (10 Marks)

Module-4

- 7 a. Explain the significance and requirements of a Highway Drainage System. (10 Marks)
b. With a neat sketch, explain lowering of water table and control of capillary rise. (10 Marks)

OR

- 8 a. Explain briefly the quantifiable and non – quantifiable benefits for a highway user. (10 Marks)
b. Write a note on : i) BOT ii) BOOT iii) VOC. (10 Marks)

Module-5

- 9 a. In detail explain the road user characteristics. (10 Marks)
b. Discuss the various traffic studies and what are the objectives of carrying out traffic volume studies. (10 Marks)

OR

- 10 a. Define Permanent way. Draw the neat diagram of cross – sectional view of permanent way and describe the component parts. (10 Marks)
b. Draw a neat sketch of layout of an airport and explain the functions of each component. (10 Marks)

* * * * *

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

21CV53

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Design of RC Structural Elements

Time: 3 hrs.

Max. Marks:100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Used of US 456 – 200, SP 16 is permitted
3. Missing data, if any, may be suitably assumed.

Module-1

- 1 a. Distinguish between working stresses and limit state methods design. (08 Marks)
b. Write a brief note on :
i) Balanced section
ii) Under reinforced section
iii) Deflection
iv) Cracking. (12 Marks)

OR

- 2 a. Define long-term and short-time deflection. What are the factors influencing the long-term and short-term. (08 Marks)
b. A simply supported beam of rectangular section spans over 8m and has an effective depth of 600 mm. The beam is reinforced with 1% reinforcement on the tension side. Check for the deflection control of the beam by empirical method if :
i) Fe – 250 grade steel
ii) Fe – 415 grade steel
iii) Fe – 500 grade steel. (12 Marks)

Module-2

- 3 What are the steps involved in the analysis of design, when the following area given, A_{st} or number of bars with diameter(ϕ) of bars, size of beam, grade of concrete, and steel, if load to be calculated then span is given. (20 Marks)

OR

- 4 a. Find the depth of neutral axis of a singly reinforced RC bema of 250 mm width and 500 mm effective depth. It is reinforced with 4 bars of 20 mm diameter. Use M20 concrete and FE 415 bars. Also check for type of section. (10 Marks)
b. A singly reinforced concrete beam 250 mm width is reinforced 4 bars of 25 mm diameter at an effective depth of 400 mm. If M20 grade concrete and Fe 415 bars are used, compute ultimate moment of resistance of the section. (10 Marks)

Module-3

- 5 List and enumerate general specification for flexure design of beams. (20 Marks)

OR

- 6 Design the T-beam as per IS : 456 – 2000. The beam is subjected to an ultimate moment of 400 kN-m. Use M20 concrete and Fe 415 steel. Following are the parameters which are used for design. $b_f = 800$ mm, $b_w = 200$ mm, $D_f = 100$ mm, $d = 400$ mm. (20 Marks)

Module-4

- 7 Design a simply supported slab on masonry walls to the following requirements. Draw plan and section showing reinforcement details.
Clear span = 2.5 m
Live load = 300 N/m^2
Use M-20 concrete and Fe-415 steel. (20 Marks)

OR

- 8 Design a dog-legged staircase for a building in which the vertical distance between floors is 3.6m. The stair hall measures $3\text{m} \times 6\text{m}$. Take live load on the stairs is 4 kN/m^2 . The flight are supported on 230 mm walls at the ends of outer edges. So, that it spans in the direction of going. Adopt M20 concrete and Fe-415 grade steel. Sketch the reinforcement details. (20 Marks)

Module-5

- 9 a. Distinguish between short column and long column. (10 Marks)
b. What are the assumptions made for the limit state of collapse in compression? (10 Marks)

OR

- 10 Design on RCC short square column to the following particulars.
Axial load = 1200 kN
Grade of concrete = M-20
Length of column = 1.85m
Grade of steel = Fe-415
Sketch the reinforcement details. (20 Marks)

* * * * *

CBCS SCHEME

USN

--	--	--	--	--	--	--	--

21CV54

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Geotechnical Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following with the help of three phase diagram :
i) Specific gravity
ii) Air content
iii) Degree of saturation
iv) Voids ratio
v) Water content. (10 Marks)
- b. With usual notations prove that $eS = wG$. (04 Marks)
- c. An undisturbed sample of soil has volume of 100cm^3 and mass of 100 g. On oven drying for 24 hours, the mass is reduced to 160 g, if the specific gravity of grains is 2.68. Determine the water content, voids ratio and degree of saturation of the soil. (06 Marks)

OR

- 2 a. Explain determination of in situ – density by sand replacement method. (10 Marks)
- b. Explain soil classification according to BIS classification. (06 Marks)
- c. Explain the particle size distribution curve. (04 Marks)

Module-2

- 3 a. Explain factors affecting permeability of soil. (10 Marks)
- b. With neat sketch explain constant head permeability and falling head permeability test. (10 Marks)

OR

- 4 a. Explain permeability of stratified soil deposits for both the cases. (10 Marks)
- b. Write short notes on following :
i) Effective stress
ii) Total stress
iii) Neutral stress. (06 Marks)
- c. In a falling head permeameter test, the initial head ($t = 0$) is 40 cm. The head drops by 5 cm in 10 minutes. Calculate the time required to run the test for the final head to be at 20 cm. If the sample is 6 cm in height and 50 cm^2 in cross – sectional area, Calculate the co-efficient of permeability, taking area of stand pipe is 0.5 cm^2 . (04 Marks)

Module-3

- 5 a. Explain factors affecting compaction of soil. (10 Marks)
- b. With neat sketch explain standard proctor test. (10 Marks)

OR

- 6 a. Explain mass spring analogy. (10 Marks)
b. With a neat sketch explain laboratory consolidation test. (10 Marks)

Module-4

- 7 a. With a neat sketch explain direct shear test. (10 Marks)
b. Explain Mohr – coulomb failure theory. (10 Marks)

OR

- 8 a. With a neat sketch explain Triaxial compression test. (10 Marks)
b. Explain factors affecting shear strength of soils. (05 Marks)
c. Explain Thixotrophy and sensitivity. (05 Marks)

Module-5

- 9 a. Explain the assumptions of Terzaghi's analysis for bearing capacity of soil. (07 Marks)
b. Explain effect of water table on bearing capacity of soil (08 Marks)
c. Explain factors affecting bearing capacity of soil. (05 Marks)

OR

- 10 a. Explain types of settlements. (10 Marks)
b. Estimate the immediate settlement of a concrete footing $1.5\text{m} \times 1.5\text{m}$ in size founded at a depth of 1 m in silty soil whose modulus of elasticity is 90 kg/cm^2 . The footing is expected to transmit a unit pressure of 200 kN/m^2 . (05 Marks)
c. Explain effect of eccentricity of loading for bearing capacity of soil. (05 Marks)

CBCS SCHEME

21CIV57

USN

--	--	--	--	--	--	--	--	--	--

Question Paper Version : A

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Environmental Studies

Time: 1 hrs.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Environment means
 - a) A beautiful landscape
 - b) Industrial production
 - c) Sum total of all condition
 - d) None of these
 2. The term ecosystem was introduced by
 - a) Hackel
 - b) Odum
 - c) Tansley
 - d) All of these
 3. The ecological pyramid that is always upright.
 - a) Pyramid of energy
 - b) Pyramid of biomass
 - c) Pyramid of number
 - d) None of these
 4. 5th June is observed as
 - a) World Forest Day
 - b) World Environment Day
 - c) World Wildlife Day
 - d) World Population Day
 5. MoEF means
 - a) Ministry of Forest and Energy
 - b) Ministry of Environment and Forests
 - c) Ministry of Fuel and Energy
 - d) Management of Environment and Forestry
 6. In Aquatic Ecosystem Phytoplankton can be considered as a
 - a) Consumer
 - b) Producer
 - c) Macro consumer
 - d) Micro consumer
 7. _____ has Maximum Genetic Diversity in India.
 - a) Potato
 - b) Tea
 - c) Mango
 - d) Teak

8. Chipko movement was started to conserve
a) Forests b) Grasslands c) Deserts d) Soil
9. The sequence of eating and being eaten in an eco-system is called
a) Food chain b) Carbon cycle
c) Hydrological cycle d) Anthroposystem
10. Which of the following is a biotic component of an eco system?
a) Fungi b) Solar Light c) Temperature d) Humidity
11. Hydroelectric Power plant is _____
a) Non-renewable source of energy b) Conventional source of energy
c) Non-conventional source of energy d) Continuous source of energy
12. Which isotope of uranium is used for the nuclear fission reaction?
a) U-234 b) U-235 c) U-238 d) U-233
13. A solar cell is an electrical device that converts the energy of light directly into electricity by the
a) Photovoltaic effect b) Chemical effect
c) Atmospheric effect d) Physical effect
14. Bhopal gas tragedy occurred in the year
a) 1986 b) 1990 c) 1984 d) 1991
15. Wind is beneficial resource of energy as it does not cause
a) Pollution b) Echo c) Noise d) Sound
16. Which of the following is not a renewable source of energy?
a) Wind energy b) Tidal wave energy
c) Solar energy d) Fossil fuels
17. Nuclear Power Plant in Karnataka is located at
a) Bhadravathi b) Kaiga c) Sandur d) Raichur
18. In Hydro Power Plants, power is generated by
a) Hot Springs b) Wind c) Water d) Solar Energy
19. Which is not natural disaster?
a) Cyclone b) Nuclear explosion
c) Earthquake d) Volcano
20. Radiation is a health hazard because it leads to
a) Typhoid b) Cancer c) Colour blindness d) Pneumonia
21. Water pollution is caused by
a) Sewage b) Industrial effluents
c) Discharge from farms d) All of these
22. Which of the following are non-biodegradable?
a) Plastics b) Domestic sewage
c) Detergents d) Both a and c

23. Chlorine can be used
 a) To kill pathogenic micro organisms b) To increase the pH
 c) To clear the turbidity d) All of these
24. What does E-waste stand for?
 a) Environment waste b) Electronic waste
 c) Equipment waste d) None of these
25. The noise is measured in
 a) Decibels b) Joules c) PPM d) NTU
26. Maximum dissolved oxygen is required by
 a) Fish b) Bacteria c) Vegetables d) All of these
27. Colorless, odorless and non corrosive air pollutant is
 a) Sulphur dioxide b) Carbon monoxide
 c) Carbon dioxide d) Ozone
28. Which of the following is not a greenhouse gas?
 a) CO₂ b) CH₄ c) CFC d) H₂
29. For the survival of fish in a river stream, the minimum DO is prescribed
 a) 3 PPM b) 4 PPM c) 5 PPM d) 10 PPM
30. Water pollution can be minimized by
 a) Releasing sewage to ocean b) Releasing effluent to wasteland
 c) Treating wastewater d) None of these
31. Global warming could effect
 a) Climate b) Increase in sea level
 c) Melting of glaciers d) All of these
32. The primary cause of acid rain around the world is
 a) Carbon dioxide b) Sulphur dioxide
 c) Carbon monoxide d) Ozone
33. Acid rain effects on
 a) Materials b) Plants c) Soil d) All of these
34. Ozone layer is present in
 a) Troposphere b) Stratosphere
 c) Mesosphere d) Thermosphere
35. Ozone layer absorbs
 a) UV rays b) Infrared rays c) Cosmic rays d) CO
36. The Fluoride concentration for prevention of dental caries is
 a) 3 mg/L b) 2 mg/L c) 1 mg/L d) 4 mg/L
37. D.D.T is a
 a) Fungicide b) Pesticide c) Fertilizer d) Disinfectant

38. When trees are cut, amount of oxygen
a) decreases b) increases c) both a and b d) remains same
39. World ozone day is being celebrated on
a) September 5th b) October 15th c) September 16th d) September 11th
40. The effect of acid rain
a) Reduces soil fertility b) Increases atmospheric temperature
c) Skin cancer d) Causing respiratory problems
41. Which among the following is not related to GIS software?
a) CAD b) ARC GIS c) RC VIEW d) STAAD PRO
42. GIS stands for
a) Geographic Information System b) Generic Information System
c) Geological Information System d) Geographic Information System
43. Among the following _____ can be expressed as an example of hardware component?
a) Keyboard b) Arc GIS c) Autocad d) Digitalization
44. The basic requirement of any sensor system is
a) Spatial resolution b) Spectral Resolution
c) Radiometric Resolution d) All of these
45. IS 14000 standards are for the
a) Quality Management System b) Environmental Management System
c) Administration d) Supply Chain
46. What is the main objective of secondary treatment of sewage plants?
a) To remove the suspended particles
b) To remove the contaminants
c) To remove the BOD
d) To remove the organic material
47. Which of the following is the un agency on health?
a) WHO b) FAO c) UNESCO d) WTO
48. What is the full form of NGO's?
a) Non-governmental Organizations b) Non Governance Organizations
c) No Governance Organizations d) Null Governmental Organizations
49. When did green peace founded?
a) 1965 b) 1967 c) 1968 d) 1971
50. When did the Bombay Natural History Society (BNHS) founded?
a) 1857 b) 1868 c) 1883 d) 1897

* * * * *

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

21CV646

Sixth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Alternative Building Materials

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss the concept of embodied energy in building materials. (10 Marks)
- b. Explain any four environmental implications caused by building materials. (10 Marks)

OR

- 2 a. Discuss recycling of any two building wastes. (10 Marks)
- b. Explain the various Biomass resources for building. (10 Marks)

Module-2

- 3 a. Discuss the raw material used for stabilized mud block and explain its manufacturing process. (08 Marks)
- b. Discuss how the following factors affect the compressive strength of masonry:
(i) Strength of masonry unit (ii) Moisture absorption (06 Marks)
- c. A solid concrete block prism consists of 4 concrete blocks of size 160×80×80 mm and the joint thickness is 2 cm. The modulus of block and mortar are 10,000 MPa and 600 MPa respectively. The Poisson's ratio of block is 0.1 and that of mortar is 0.15. Determine the magnitude and nature of lateral stress in block and mortar when vertical stress is 6.0 MPa. (06 Marks)

OR

- 4 a. Write a note on :
(i) Laterite block (ii) Hollow Clay Block. (08 Marks)
- b. Discuss the factors to be considered for selection of mortar. (06 Marks)
- c. Explain the following briefly: (i) Masonry prism (ii) Masonry wallets (06 Marks)

Module-3

- 5 a. With a neat sketch discuss the manufacturing process of lime. (10 Marks)
- b. Discuss the application of following industrial waste in building construction.
(i) Fly ash (ii) Blast furnace slag (iii) Marble Dust (iv) Demolished building waste (10 Marks)

OR

- 6 a. Discuss the various types of fibres that can be used in fibre reinforced plastics. (10 Marks)
- b. Discuss the manufacturing process of Dry blending and wet blending of Lime pozzolana cement mortar. (10 Marks)

Module-4

- 7 a. Discuss the following with neat sketches:
(i) Rammed Earth Wall (ii) Cavity Wall (10 Marks)
- b. With the help of neat sketch, explain the following :
(i) Filler slab (ii) Composite beam panel roof. (10 Marks)

OR

- 8 a. Explain the concept of ferrocement. Also discuss the raw materials used. (10 Marks)
b. Discuss the following :
(i) Mivan construction technique
(ii) Top down construction (10 Marks)

Module-5

- 9 a. Discuss the machines used in following stages of concrete:
(i) Mixing
(ii) Transportation of concrete. (10 Marks)
b. Explain briefly about the cost saving technique in planning, design and construction. (10 Marks)

OR

- 10 a. List the various equipments used for production of stabilized mud block and explain any one in brief. (10 Marks)
b. Write a note on precast elements and discuss its advantages and disadvantages. (10 Marks)

* * * * *

--	--	--	--	--	--	--	--	--	--

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Quantity Survey and Contract Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

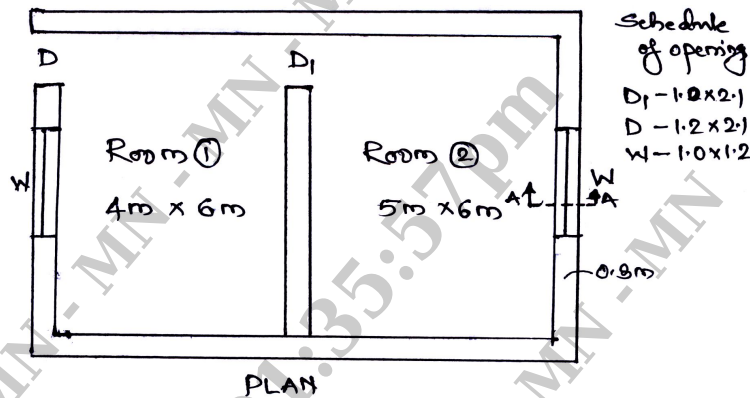
Module-1

- 1 Define estimate. Explain briefly purpose and different types of estimate (any three). (20 Marks)

OR

- 2 The details of two room building are shown in Fig.Q.2. Estimate the quantities and cost of the following items of work:

- Earthwork excavation for foundation in ordinary soil @ Rs.400/m³.
- Cement concrete bed 1:4:8 for wall foundation @ Rs.3800/m³.
- SSM in CM 1:6 for foundation and basement @ Rs.3000/m³.
- Ist class BBM work for super structure in CM 1:6 @ Rs.5800/m³.
- RCC 1:1.5:3 roof slab at Rs.5500/m³.



Schedule of opening
 D₁ - 1.2 x 2.1
 D - 1.2 x 2.1
 W - 1.0 x 1.2

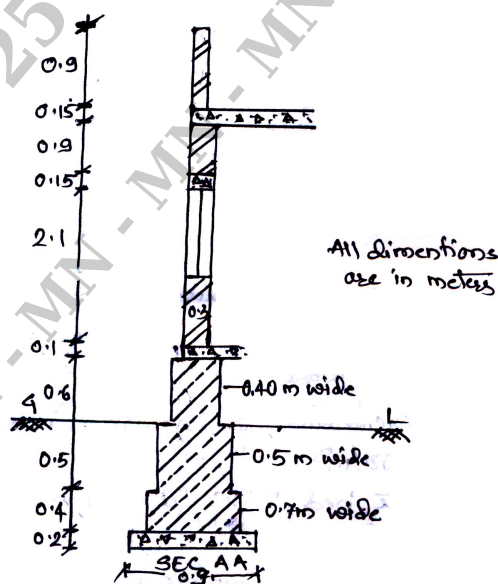


Fig.Q.2

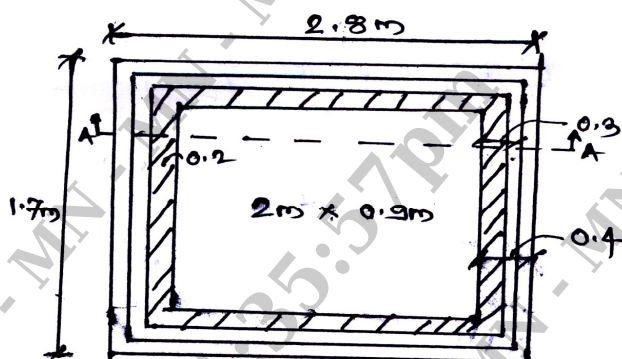
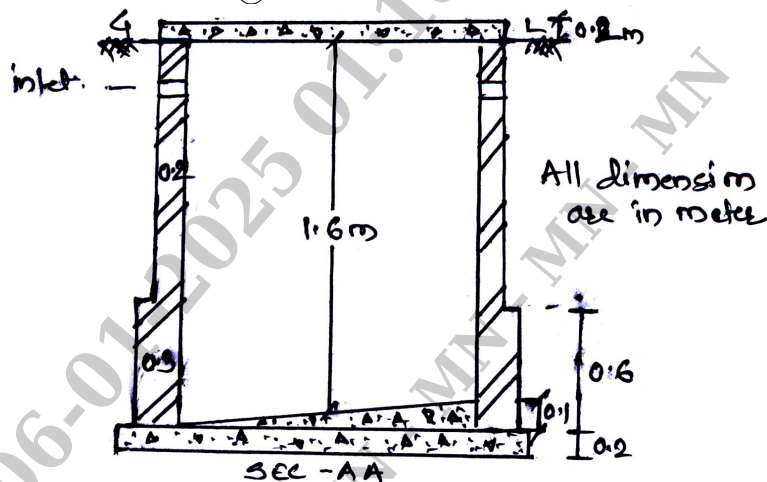
1 of 3

(20 Marks)

Module-2

3 The details of a septic tank is as shown in Fig.Q.3. Estimate the quantities for the following items of work and work out the cost.

- Earth work excavation in foundation @ Rs.400/m³
- Cement concrete 1:3:6 floor and foundation @ Rs.4000/m³
- First class brick work with cement mortar 1:4 @ Rs.6000/m³.
- 12 mm thick plaster for side walls @ Rs.200/m².



PLAN.
Fig.Q.3

(20 Marks)

OR

4 Reduced level (RL of ground along centerline of proposed road from chainage 10 to 20 chainage are given below. The formation level at the 10th chainage is 107 and the road is in down word gradient of 1 in 150 upto the chainage 14 and the gradient changes to 1 in 100 down word gradient. Formation width is 10m and side slope of banking are 2:1 (H:V). Length of chain is 30 m. Estimate the quantities and cost of earth at the rate for filling is Rs.250 and cutting Rs.140 by mid sec. method.

Chainage	10	11	12	13	14	15	16	17	18	19	20
RL of the GL	105.0	105.60	105.44	105.90	105.42	104.30	105.00	104.10	104.62	104.00	103.30
RL of FL	107.00										
Gradient	Down word 1 in 150					Down word 1 in 100					

(20 Marks)

Module-3

- 5 a. Define specification. Explain the objectives of writing specification. (10 Marks)
 b. List the significances of microsoft excel in estimation. (10 Marks)

OR

- 6 Write detailed specification for following: (20 Marks)
 i) Bed concrete for foundation with M7.5 CC
 ii) Size stone masonry for foundation in CM 1:6
 iii) Plastering 12 mm thick with CM 1:5
 iv) Floor filling with granite in CM 1:3

Module-4

- 7 a. Define rate analysis. Explain the purpose of rate analysis. (10 Marks)
 b. Work out the rate analysis for R.C.C. for slab and beam in CC 1:1.5:3 and steel 1.2%. (10 Marks)

OR

- 8 Analyze rates from first principle for following: (20 Marks)
 i) Ashlar masonry for foundation in CM 1:6
 ii) BBM for super structure in CM 1:6
 iii) Plastering 20 mm thick in CM 1:4
 iv) Pointing in CM 1:3.

Module-5

- 9 a. Define contracts. Discuss the essentials of contracts. (10 Marks)
 b. Write a note on tender documents and earnest money deposit. (10 Marks)

OR

- 10 Write short notes on following: (20 Marks)
 a. Administrative approval and technical sanction
 b. Nominal muster roll
 c. Security deposit
 d. Running bills.

* * * * *

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

21CV72

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Construction Technology for Substructures and Super Structures

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Write a note on Micro Tunnelling. (10 Marks)
- b. Write a note on TBM. (10 Marks)

OR

- 2 a. What is bedding of conduits explain its significance. (10 Marks)
- b. Explain different methods of compaction technology. (10 Marks)

Module-2

- 3 a. What is underwater drilling and blasting give one case study. (10 Marks)
- b. Explain Jet grouting with a neat sketch. (10 Marks)

OR

- 4 a. What is the role of dewatering in shallow and deep excavation and list various method. (10 Marks)
- b. Explain well point system with a neat sketch. (10 Marks)

Module-3

- 5 a. Write a note on operation and erection of R.M.C plant. (10 Marks)
- b. Write a note on Concrete mix design. (10 Marks)

OR

- 6 a. Explain underwater concrete and its significance and also list the various methods used. (10 Marks)
- b. Write a note on Self-Compacting Concrete. (10 Marks)

Module-4

- 7 a. Write a note on piles in land and marine structures. (10 Marks)
- b. Explain initial and routine tests on piles as per codal specifications. (10 Marks)

OR

- 8 a. Write a note on pile driving by vibration and construction of micro piles. (10 Marks)
- b. Write a note on Diaphragm walls. (10 Marks)

Module-5

- 9 a. What are cofferdams and explain the construction of a double wall cofferdam? (10 Marks)
- b. State the uses of cofferdams and explain concrete wall movable cofferdams. (10 Marks)

OR

- 10 a. What are the differences between pneumatic and open caissons? (10 Marks)
- b. Write a note on Well foundations. (10 Marks)

* * * * *

Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

21CV731

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Advanced Design of RCC and Steel Structures

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any TWO full questions, choosing ONE full question from each module.
2. Use of IS456, IS-800, SP(6) and Steel tables are permitted.

Module-1

- 1 Design a combined footing for two columns of size 300×300 mm and $400 \text{ mm} \times 400 \text{ mm}$ subjected to 500 kN and 700 kN respectively. The C/C spacing between the columns is 3.5 m. The width of the footing is restricted to 1.5 m. Take SBC of the soil as 150 kN/m^2 . Use M_{25} concrete and Fe_{415} steel. Design slab and beam type combined footing. Draw the sectional elevations. (50 Marks)

OR

- 2 Design a cantilever retaining wall to retain on Earth embankment 4 m high above the ground level. The density of the earth is 18 kN/m^3 and its angle of repose is 30° . The embankment is horizontal at its top. The SBC of the soil may be taken as 200 kN/m^2 and the co-efficient of friction between soil and concrete is 0.5. Adopt M_{20} grade concrete and Fe_{415} steel. Draw cross sectional elevation showing reinforcement details. (50 Marks)

Module-2

- 3 A line diagram of Roof truss with internal loads and forces in each member are shown in Fig.Q3. The design of various member of the roof truss along with their end connection with bolt of property class 5.6. Also, design the bearing plate at support for the reaction and Anchor bolts for an uplift force of 15 kN. Also draw the elevation of the truss greater than half span.

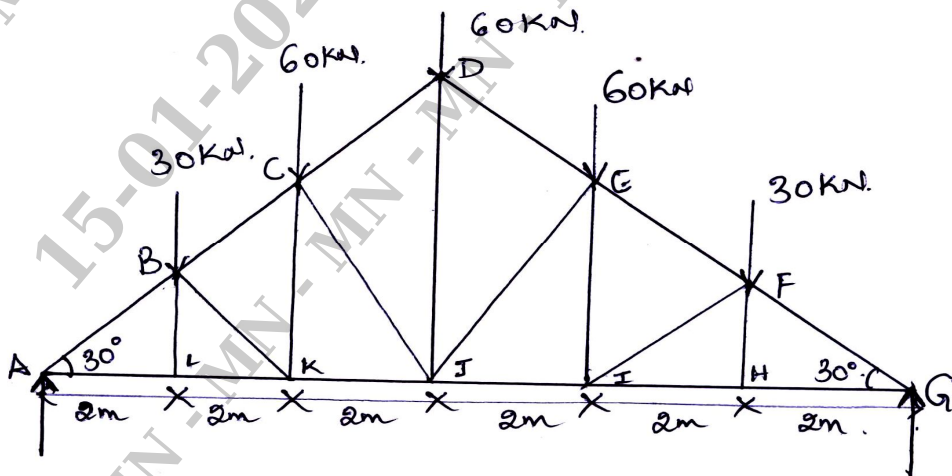


Fig.Q3

Members	Length (m)	Force (kN)	Nature of force
AB, GF	2.31	240	C
BC, FE	2.31	210	C
CD, ED	2.31	160.04	C
AL, GH	2.0	207.84	T
LK, HI	2.0	207.84	T
KJ, IJ	2.0	181.82	T
BL, FH	1.154	0	-
BK, FI	2.31	30	C
CK, EI	2.31	15	T
CJ, EJ	3.05	66	C
DJ	3.46	60	C

(50 Marks)

OR

- 4 Design a simply supported Gantry Girder for the following details. The Girder is electrically operated. Take yield stress of the steel as 250 MPa.
- Span of Crane Girder = 20 m
 - Span of Gantry Girder = 7 m
 - Wheel base distance = 3.4 m
 - Self weight of crane excluding crab = 200 kN
 - Weight of trolley = 60 kN
 - Min hook approach = 1.1 m
 - Self weight of rail = 0.3 kN/m
 - Height of rail = 75 mm
 - Capacity of crane = 250 kN

(50 Marks)

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

21CV742

Seventh Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Ground Improvement Technique

Time: 3 hrs.

Max. Marks: 100

Note : 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Assume missing data if any.

Module-1

- 1 a. Discuss about effect of compaction on Engineering properties of Soil. (10 Marks)
- b. Explain about construction of stone columns for deep compaction with neat sketch. (10 Marks)

OR

- 2 a. Explain how proctor needle is used to control compaction in field. (10 Marks)
- b. A proposed embankment fill required 12,000 m³ of compacted soil. The void ratio of compacted fill is specified as 0.65. The following table list 4 available borrow pits along with void ratio and cost per cum of moving the soil to proposed site. Assuming $G = 2.7$. Show the suitability of each site.

Borrow pit	Void ratio	Cost in Rs/m ³
A	0.9	60
B	1.1	50
C	0.95	80
D	0.75	10

(10 Marks)

Module-2

- 3 a. Explain about the Mechanism of soil – lime stabilization technique. (10 Marks)
- b. Discuss about Engineering behavior of soil – lime stabilization. (10 Marks)

OR

- 4 a. List the Engineering properties of fly ash. (10 Marks)
- b. Outline about change in properties of soil by calcium chloride stabilization. (10 Marks)

Module-3

- 5 a. Air mark the main object of Dewatering technique in soil. (10 Marks)
- b. Explain with neat sketch about Electro – Osmosis method of dewatering. (10 Marks)

OR

- 6 a. Write a note on seepage control in construction site. (10 Marks)
- b. List the various methods of dewatering. Explain any one method. (10 Marks)

Module-4

- 7 a. Discuss about the components of Reinforced Earth with typical sketch. (10 Marks)
- b. List the advantages of Reinforced Earth structure and also highlight about Application of soil reinforcement. (10 Marks)

OR

- 8 a. Explain step by step procedure of construction of soil nailing technique. (10 Marks)
b. List the advantages and disadvantages of soil nailing. (10 Marks)

Module-5

- 9 a. List the classification of Geotextiles. (10 Marks)
b. Explain about the applications of Geosynthetics. (10 Marks)

OR

- 10 a. Write a note on Anchor Rock bolts. (10 Marks)
b. Explain construction and advantages of Micropile with supporting sketch. (10 Marks)

* * * * *

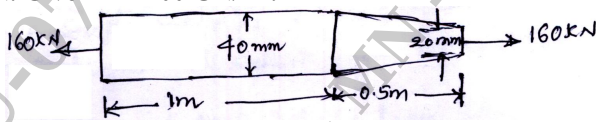
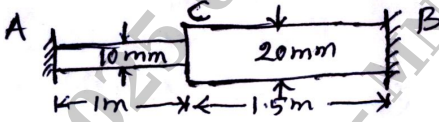
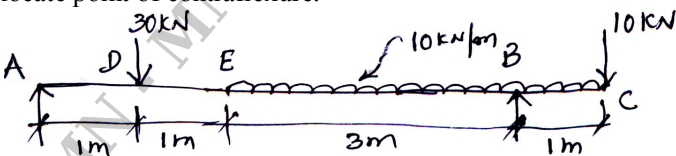
Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Strength of Materials

Time: 3 hrs.

Max. Marks: 100

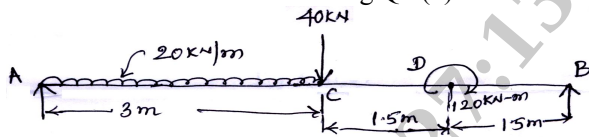
Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. *M* : Marks , *L*: Bloom's level , *C*: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Sketch a typical stress-strain curve for a ductile material and explain briefly the salient features of the curve.	6	L2	CO1
	b.	Define: i) Strain ii) Poisson's ratio.	4	L1	CO1
	c.	A 1.5 m long steel bar is having uniform diameter of 40 mm for a length of 1 m. In the next 0.5 m its diameter gradually reduces from 40 mm to 20 mm as shown in Fig.Q.1(c). Determine the elongation of this bar when subjected to an axial tensile load of 160 kN. Given $E = 200 \text{ GN/m}^2$.	10	L3	CO1
 Fig.Q.1(c)					
OR					
Q.2	a.	What are the elastic constants and explain them briefly?	6	L2	CO1
	b.	Explain St Venant's principle.	4	L2	CO1
	c.	The steel rod shown in Fig.Q.2(c) is in two parts. It has a diameter of 10 mm for a length of 1 m and 20 mm for the remaining length of 1.5 m. If it is constrained between two parts A and B and is stress free at 20°C. Find the stress in the material, when it is subjected to 70°C. $E = 200 \text{ GPa}$, $\alpha = 12 \times 10^{-6}/^\circ\text{C}$.	10	L3	CO1
 Fig.Q.2(c)					
Module – 2					
Q.3	a.	For a cantilever beam subjected to a udl of intensity w /unit length throughout, plot SFD and BMD.	6	L3	CO2
	b.	For a simply supported overhanging beam as shown in Fig.Q.3(b), draw SFD and BMD and locate point of contraflexure.	14	L3	CO2
 Fig.Q.3(b)					

1 of 3

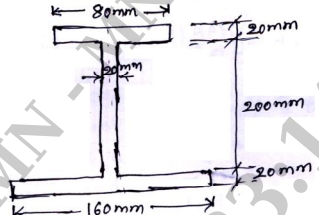
OR

Q.4	a.	For a simply supported beam subjected to udl of w /unit length throughout. Plot SFD and BMD.	6	L3	CO2
	b.	Draw SFD and BMD for the beam shown in Fig.Q.4(b).  Fig.Q.4(b)	14	L3	CO2

Module – 3

Q.5	a.	Derive the relation between bending stress and radius of curvature $\frac{\sigma}{y} = \frac{E}{R}$.	6	L3	CO3
	b.	Define: i) Neutral axis ii) Section modulus	4	L2	CO3
	c.	A hollow propeller shaft of a steam ship is to transmit 3750 KW at 240 rpm. If the internal diameter is 0.8 times the external diameter and if the maximum shear stress developed is to be limited to 160 N/mm ² , determine the size of the shaft.	10	L3	CO4

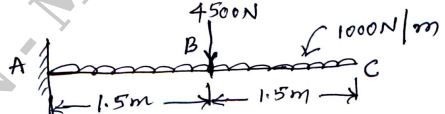
OR

Q.6	a.	Derive the torsion equation with usual notations.	8	L3	CO4
	b.	The unsymmetrical I-section shown in Fig.Q.6(b) is subjected to a shear force of 40 KN. Draw the shear stress variation diagram across the depth.  Fig.Q.6(b)	12	L3	CO3

Module – 4

Q.7	a.	Derive differential equation for beam deflection with usual notations.	8	L3	CO4
	b.	Determine the critical load for a hollow cast iron rectangular column of external dimensions 200 mm × 150 mm with the thickness of the metal being 25 mm. The height of the column is 6 m and both ends are fixed. Use Euler's formula and compare the value with that obtained by using Rankine's formula taking $\sigma_c = 500$ N/mm ² and $a = 1/1600$, which of the above formula decides the safe crippling load?	12	L3	CO4

OR

Q.8	a.	Derive an expression for Euler's crippling load for both ends hinged columns with usual notations.	8	L3	CO4
	b.	Compute the mid-span and maximum deflection for the beam shown in Fig.Q.8(b). Given $E = 210$ GN/m ² and moment of inertia = 36000×10^{-9} m ⁴ .  Fig.Q.8(b)	12	L3	CO4

Module – 5

Q.9	a.	With a neat sketch, explain the two-dimensional stress system.	8	L2	CO5
	b.	A cylindrical shell is 3 m long and is having 1 m internal diameter and 15 mm thickness. Calculate the maximum intensity of shear stress induced and also the changes in the dimensions of the shell, if it is subjected to an internal fluid pressure of 1.5 N/mm^2 .	12	L3	CO5
OR					
Q.10	a.	Derive Lamé's equation for radial and hoop stress for thick cylinder subjected to internal and external fluid pressure.	10	L3	CO5
	b.	The direct stresses acting at a point in a strained material are shown in Fig.Q.10(b). Find the normal, tangential and resultant stresses on plane 30° to the plane of major principal stress. Find the obliquity of the resultant stress also.	10	L3	CO5

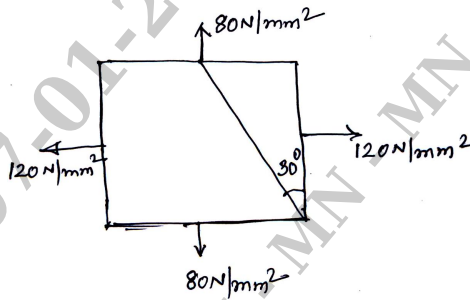


Fig.Q.10(b)

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

BCV302

Third Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025 Engineering Survey

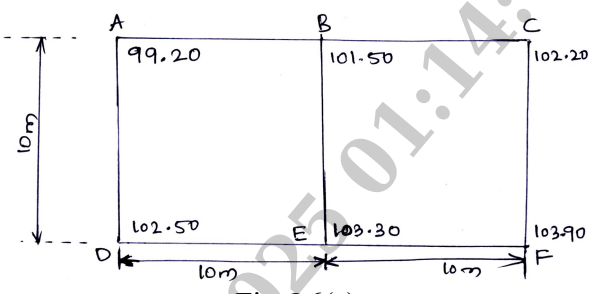
Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
1	a.	What is Surveying? Explain the principles of surveying.	8	L2	CO1
	b.	Compare between : i) True and magnetic bearing ii) Plane and geodetic surveying iii) Cadastral and hydrographic surveying.	6	L1	CO1
	c.	Write the advantages and disadvantages of plane table surveying.	6	L1	CO1
OR					
2	a.	Discuss on importance of surveying in civil engineering.	6	L2	CO1
	b.	Explain the different methods of measuring distances with examples.	6	L2	CO1
	c.	Differentiate between : i) Chain and chainage ii) Laser distance meter and distance measuring wheel iii) EDM and GPS iv) Topographical and construction survey.	8	L1	CO1
Module – 2					
3	a.	The following consecutive readings were taken with a level and a 4m staff on a continuously sloping ground at a common interval of 20m : 0.780, 1.535, 1.955, 2.430, 2.985, 3.480, 1.155, 1.960, 2.365, 3.640, 0.935, 1.045, 1.630 and 2.545. The RL of first point A was 180.750m. Rule out a page of level field book and enter the above readings. Compute the RL's by HI method. Also calculate the gradient of the line joining the first and last points.	10	L3	CO2
	b.	List the salient features of total station.	6	L2	CO2
	c.	Why fly and flyback leveling is required?	4	L2	CO2
OR					
4	a.	Explain the temporary adjustments of dumpy level.	6	L2	CO2
	b.	Explain the method of measuring horizontal angle by repetition method along with the tabular column.	10	L3	CO2
	c.	Discuss on different fundamental measurements of total station.	4	L1	CO2
Module – 3					
5	a.	What are contours? Explain the characteristics of contours with neat sketches.	10	L3	CO3
	b.	Brief on longitudinal and cross-sectioning with typical sketches.	8	L2	CO3
	c.	What are the input data required while creating job file in total station?	2	L1	CO3

OR

6	a.	Plot the contours of RL 100.00, 101.00, 102.00, and 103.00 in the given square blocks of 10 m × 10 m. The reduced levels of guide points are given in Fig.Q6(a).	10	L3	CO3
		 Fig.Q6(a)			
	b.	Explain the following related to total station : i) Back sight data ii) Coordinates data iii) Command to plot contours in auto CAD. iv) Data transferring.	4	L2	CO3
	c.	Explain the procedure and select the contour interval.	6	L2	CO3

Module – 4

7	a.	Calculate the necessary data for setting out simple curve of radius 300m with the deflection angle of 50° 30'. The two tangents intersect at a chainage of 1192.00m. Take peg interval as 20m and tabulate the results using Rankine's method.	6	L3	CO4																		
	b.	List the different types of curves.	4	L1	CO4																		
	c.	The following perpendicular offsets were taken from chain line to an irregular boundary. Calculate the area enclosed by trapezoidal rule. <table border="1"><tr><td>Chainage (m)</td><td>0</td><td>30</td><td>60</td><td>90</td><td>120</td><td>150</td><td>180</td><td>210</td></tr><tr><td>Offset (m)</td><td>0</td><td>2.65</td><td>3.80</td><td>3.75</td><td>4.65</td><td>3.60</td><td>5.0</td><td>5.80</td></tr></table>	Chainage (m)	0	30	60	90	120	150	180	210	Offset (m)	0	2.65	3.80	3.75	4.65	3.60	5.0	5.80	10	L2	CO4
Chainage (m)	0	30	60	90	120	150	180	210															
Offset (m)	0	2.65	3.80	3.75	4.65	3.60	5.0	5.80															

OR

8	a.	A railway embankment is 10m wide with side slopes 1.5 to 1.0. Assuming ground to be level in a direction transverse to center line, calculate the volume contained in a length of 120 meters, the centre heights at 20m intervals being are 2.2, 3.7, 3.8, 4.0, 3.8, 2.8 and 2.5m. Use both trapezoidal and prismoidal method.	10	L3	CO4
	b.	Sketch out a compound curve and show the elements of it.	4	L2	CO4
	c.	Define the following related to setting out works : i) Stake ii) Post iii) Batter – board iv) Sight rail.	6	L1	CO4

Module – 5

9	a.	Discuss on the various segments of GPS.	8	L2	CO5
	b.	List the applications of RS and GIS in civil engineering.	6	L1	CO5
	c.	List out the steps in drone surveying.	6	L1	CO5

OR

10	a.	Discuss on GPS receivers.	8	L2	CO5
	b.	List the features and applications of drone surveying.	8	L1	CO5
	c.	Name the type of sensors used in drone surveying.	4	L1	CO5

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

BCV303

Third Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025 Engineering Geology

Time: 3 hrs.

Max. Marks:100

**Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.**

Module – 1			M	L	C
1	a.	Explain importance of geology in Civil Engineering.	5	L2	CO1
	b.	Describe internal structure of earth with labelled sketch.	10	L2	CO1
	c.	What is land slide? Explain causes at landslides.	5	L1	CO1
OR					
2	a.	Discuss causes and effect of earthquake.	7	L2	CO2
	b.	What is Tsunami? Add a note on mitigation of Tsunami.	7	L3	CO3
	c.	Explain causes and types of volcanoes.	6	L2	CO2
Module – 2					
3	a.	What is mineral? Explain properties of mineral.	8	L2	CO2
	b.	Explain different types of igneous rocks.	6	L2	CO2
	c.	What is metamorphism? Discuss types of metamorphism.	6	L2	CO2
OR					
4	a.	Describe properties of following minerals : i) Orthoclase ii) Biotite iii) Haematite.	6	L2	CO2
	b.	Enumerate stages of development of sedimentary rock.	7	L2	CO2
	c.	Discuss important uses of different rock type.	7	L2	CO2
Module – 3					
5	a.	With a neat sketch explain soil profile.	7	L2	CO3
	b.	What is weathering? Explain types of weathering.	8	L2	CO3
	c.	Discuss classification of different soil type.	5	L2	CO3
OR					
6	a.	Explain classification of soil based on grain size.	7	L2	CO3
	b.	Discuss the effect of weathering on monumental rock.	7	L2	CO3
	c.	Write a note on : i) Drifted soil ii) Laterite soil.	6	L1	CO3

Module – 4

7	a.	Explain different rock deformation and causes.	8	L2	CO4
	b.	A bed of lime stone is dip 25° East and it has width of outcrop – 160 m. Determine true and vertical thickness.	6	L3	CO4
	c.	What is fault? Explain type of fault.	6	L2	CO4

OR

8	a.	Discuss feasibility of Dam site in folded and faulted region.	7	L2	CO4
	b.	A bed of shale is dipping maximum of 45° along S 60°E. Determine the amount and apparent dip along S 70°E.	6	L2	CO4
	c.	Write a note on : i) Dip and strike ii) Out crop.	7	L2	CO4

Module – 5

9	a.	Explain different water bearing formations.	8	L2	CO5
	b.	What an Aquifer? Discuss types of aquifer.	8	L2	CO5
	c.	Write a note on co-efficient of permeability.	4	L1	CO5

OR

10	a.	Explain electrical resistivity survey method in ground water exploration.	12	L2	CO5
	b.	Discuss factors affecting on permeability of rocks.	8	L2	CO5

* * * * *

BCV304

Third Semester B.E/B.Tech. Degree Examination, Dec.2024/Jan.2025
Water Supply and Wastewater Engineering

Time: 3 hrs.

Max. Marks:100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1						M	L	C												
1	a.	Briefly explain the need and importance of protected water supply system.				6	L2	CO1												
	b.	Explain the various types of water demand.				8	L2	CO1												
	c.	Describe the incremental method of estimating the population of a locality.				6	L2	CO1												
OR																				
2	a.	What is meant by per capita demand? Mention the factors that affect per capita demand.				6	L1	CO1												
	b.	Give the drinking water standards for the following parameters. Discuss their effect when they exceed their limits : i) Turbidity ii) Hardness iii) Fluoride iv) Nitrate.				8	L2	CO2												
	c.	The population data of a town are given below : <table border="1"><tr><td>Year</td><td>1990</td><td>2000</td><td>2010</td><td>2020</td><td>2030</td></tr><tr><td>Population</td><td>90,000</td><td>1,20,000</td><td>1,60,000</td><td>2,50,000</td><td>2,60,000</td></tr></table> Estimate the expected population in the year 2050 by geometric increase method.				Year	1990	2000	2010	2020	2030	Population	90,000	1,20,000	1,60,000	2,50,000	2,60,000	6	L3	CO1
Year	1990	2000	2010	2020	2030															
Population	90,000	1,20,000	1,60,000	2,50,000	2,60,000															
Module – 2																				
3	a.	Briefly explain the unit treatment processes of a water treatment plant with flow chart.				10	L2	CO3												
	b.	The maximum daily demand at a water purification plant has been estimated as 12 million liter per day. Design a rectangular sedimentation tank (fitted with mechanical sludge removal arrangements) for the raw supplies, assuming a detention period of 6 hours and the velocity of flow as 20 cm per minute.				5	L3	CO3												
	c.	Explain theory of filtration.				5	L3	CO3												
OR																				
4	a.	What is meant by aeration? Explain the different types of aerators.				6	L1	CO3												
	b.	What is coagulation? List the different types of coagulants used in water treatment plant.				4	L1	CO3												
	c.	Explain the construction and working of a rapid gravity sand filter with a neat sketch.				10	L2	CO3												
Module – 3																				
5	a.	List the minor methods of disinfection and explain any two methods in detail.				8	L1	CO3												
	b.	Explain the need for sanitation. Define the following terms : i) Sullage ii) Sewage.				6	L2	CO4												
	c.	Define sampling of water or wastewater. Explain the different methods of sampling.				6	L2	CO4												
1 of 2																				

OR

6	a.	What is softening of water? List the methods used to remove hardness of water. Explain any one method in detail.	6	L2	CO3
	b.	Define disinfection. Explain break point chlorination.	4	L2	CO3
	c.	List the different types of sewerage systems with their advantages and disadvantages.	10	L2	CO4

Module – 4

7	a.	Explain with a neat flow diagram the various treatment unit operations and process used in municipal wastewater treatment.	10	L2	CO4
	b.	What is meant by activated sludge process? Describe with sketch, the treatment of sewage by activated sludge process.	10	L2	CO4

OR

8	a.	Discuss briefly with a neat sketch grit chamber and oil and grease removal tank.	10	L2	CO4
	b.	An average operating data for conventional activated sludge treatment plant is as follows : i) Wastewater flow = 35000 m ³ /d ii) Volume of aeration tank = 10900 m ³ iii) Influent BOD = 250 mg/L iv) Effluent BOD = 20 mg/L v) Mixed liquor suspended solids (MLSS) = 2500 mg/L vi) Effluent suspended solids = 30 mg/L vii) Waste sludge suspended solids = 9700 mg/L viii) Quality of waste sludge = 220 m ³ /d Based on the given information, Determine : i) Aeration period (hrs) ii) F/M ratio iii) Percentage efficiency of BOD removal iv) Sludge age (days).	10	L3	CO4

Module – 5

9	a.	Explain with a neat sketch the construction and operation of a trickling filter.	10	L2	CO4
	b.	Explain the following : i) Rotating biological contactors ii) Aerobic and anaerobic process.	10	L2	CO5

OR

10	a.	Determine the volume depth and efficiency of a building filter unit for influent sewage of 4.5 MLD, BOD ₅ of the influent is 160 mg/L. The organic loading is to be 160 gm/m ³ /day and surface loading is 2000 l/m ² /day.	10	L3	CO4
	b.	Explain the following : i) Oxidation pond ii) Stages in sludge digestion process.	10	L2	CO5

Third Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Sustainable Design Concept for Building Services

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the concept of energy security and its importance in the global energy scenario.	10	L2	CO1
	b.	What are the different types of shading systems? What are the benefits and draw backs of each type of shading systems.	10	L1	CO1
OR					
Q.2	a.	Explain the different between carbon foot print and carbon offsetting and also explain the role of carbon offsetting in achieving net zero.	10	L2	CO1
	b.	Discuss how climate change is impacting buildings and their energy efficiency? What design strategies can be used to adapt to a changing climate?	10	L2	CO1
Module – 2					
Q.3	a.	Describe the process of conduction, convection and radiation in the context of building heat transfer.	10	L2	CO2
	b.	How do factors such as bridling materials, furnishings and occupants activities affect indoor air quality?	10	L2	CO2
OR					
Q.4	a.	Explain the importance of building acoustics in creating a comfortable and productive indoor environment.	10	L2	CO2
	b.	Explain the principles of good day lighting design.	10	L2	CO2
Module – 3					
Q.5	a.	Explain the importance of building envelope insulation in reducing energy consumption for heating and cooling.	10	L2	CO3
	b.	Explain the benefits of rain water harvesting and also explain types of rain water harvesting systems.	10	L2	CO3
OR					
Q.6	a.	Explain the key benefits, environmental advantages and economic benefits of implementing ECBC 2017.	10	L2	CO3
	b.	Explain the challenges associated with recycling and reusing construction and demolition waste.	10	L2	CO3
Module – 4					
Q.7	a.	Identify and explain the factors to be considered for using most suitable sustainable materials for a specific building.	10	L3	CO4
	b.	Evaluate the environmental impact of different transportation options based on their green house gas emissions.	10	L3	CO4
1 of 2					

OR

Q.8	a.	Identify potential challenges associated with using zero-emission bricks in a construction project.	10	L3	CO4
	b.	Identify and discuss the relationship between green house gas emissions and climate change.	10	L3	CO4

Module – 5

Q.9	a.	Analyze the strength and weakness of each of the major green building rating systems.	10	L4	CO5
	b.	Evaluate the strength of weakness of different approaches to site solution in the context of IGBC certification.	10	L4	CO5

OR

Q.10	a.	Evaluate the impact of green building rating systems on the construction industry.	10	L4	CO5
	b.	Analyze the potential benefits and challenges of different pre-design strategies for a green building project.	10	L4	CO5

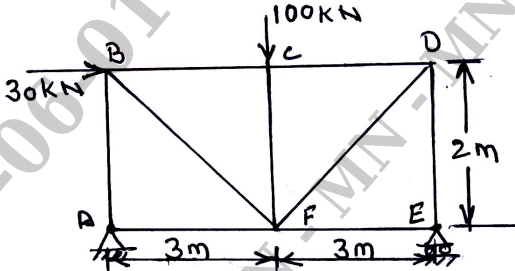
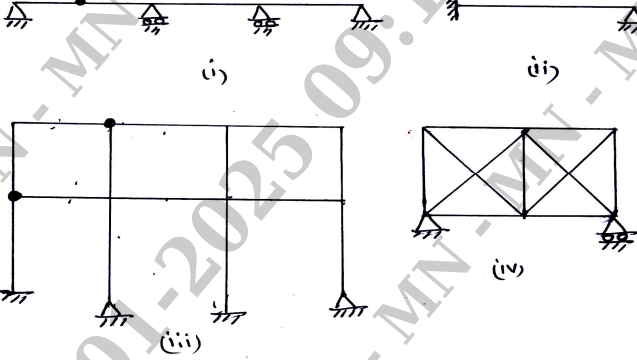
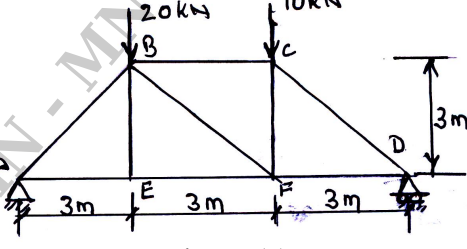
Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

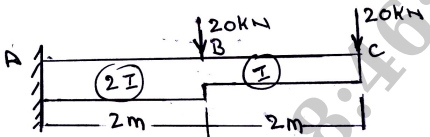
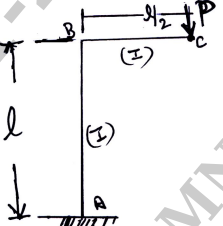
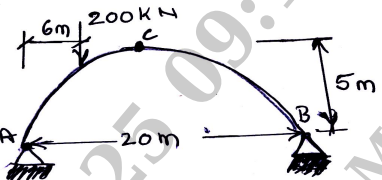
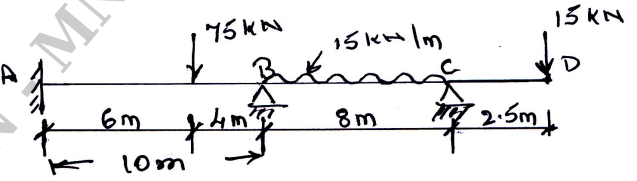
Analysis of Structures

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module - 1			M	L	C
Q.1	a.	Differentiate between statically determinate and indeterminate beams with an example for each.	08	L1	CO1
	b.	Determine the forces in all the members of the truss as shown in Fig.Q1(b). Use the method of joints.	12	L3	CO1
 Fig.Q1(b)					
OR					
Q.2	a.	Define equilibrium and compatibility conditions.	03	L1	CO1
	b.	Determine static and kinematic indeterminacy for the following shown in Fig.Q2(b).	08	L2	CO1
 Fig.Q2(b)					
	c.	Determine the support reactions and the forces in members EF, BC and BF for the truss shown in Fig.Q2(c) by method of section.	09	L3	CO1
 Fig.Q2(c)					

Module – 2					
Q.3	a.	State and explain Mohr's theorems.	06	L1	CO2
	b.	Determine the slope and deflection at free end of cantilever by using moment area method. [Refer Fig.Q3(b)]	14	L3	CO2
 Fig.Q3(b)					
OR					
Q.4	a.	Derive the expression for strain energy due to bending.	08	L1	CO2
	b.	Determine the horizontal and vertical deflection at the free end of bracket shown in Fig.Q4(b).	12	L3	CO2
 Fig.Q4(b)					
Module – 3					
Q.5	a.	Show that the parabolic shape is a funicular shape for a three hinged arch subjected to UDL over its entire span.	08	L2	CO3
	b.	A three hinged parabolic arch of span 20 m and a central rise of 5 m carry a point load of 200 kN at 6 m from the left support. Find the support reactions at A and B. Calculate normal thrust and radial shear at 6 m from the left support. Also draw the BMD. Refer Fig.Q5(b).	12	L3	CO3
 Fig.Q5(b)					
OR					
Q.6	a.	Derive the equation for cable profile and tension in the cable when it is supported at the same level and subjected to horizontal UDL.	08	L2	CO3
	b.	A cable of span 120 m and central dip 4 m carries a UDL of 20 kN/m. Determine (i) The maximum and minimum tension in the cable and its inclination (ii) Length of cable (iii) The size of cable if the permissible stress is 200 N/mm ² .	12	L3	CO3
Module – 4					
Q.7		Analyze the continuous beam shown in Fig.Q7 by slope deflection method. Draw BMD and SFD.	20	L2	CO1
 Fig.Q7					

OR

Q.8	a.	Explain fixed end moments for different loading and support conditions with relevant diagrams.	05	L1	CO4
	b.	Analyse the given frame as shown in Fig.Q8(b) by slope deflection method. EI is constant for all the members. Draw BMD and Elastic curve.	15	L4	CO4

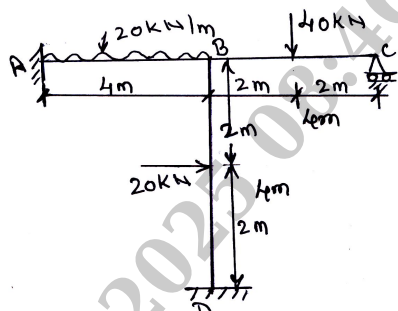


Fig.Q8(b)

Module – 5

Q.9		Analyse the continuous beam as shown in Fig.Q9 by moment distribution method and draw the BM diagram. The support B sinks by 9 mm. Take $EI = 1 \times 10^{12} \text{ N-mm}^2$.	20	L4	CO5
-----	--	--	----	----	-----

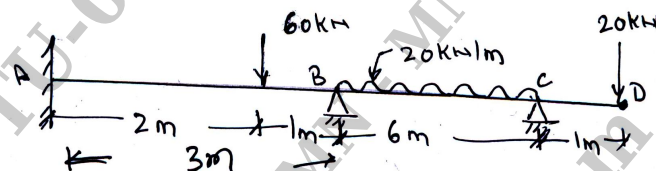


Fig.Q9

OR

Q.10		Analyse the frame shown in Fig.Q10 by moment distribution method and draw the BMD. Assume EI constant.	20	L4	CO5
------	--	--	----	----	-----

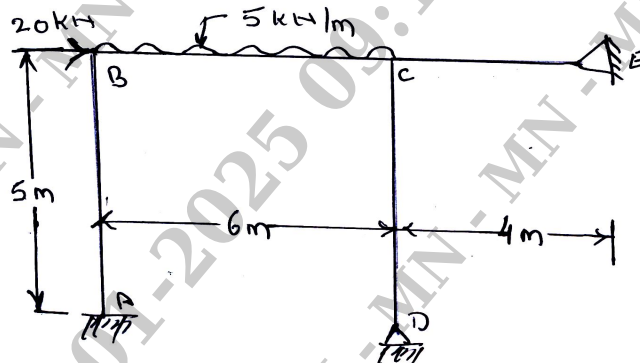


Fig.Q10

--	--	--	--	--	--	--	--	--	--

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Fluid Mechanics and Hydraulics

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define the following terms along with symbols and units: (i) Compressibility (ii) Mass density (iii) Specific weight (iv) Relative density (v) Surface tension	10	L1	CO1
	b.	A differential manometer is connected at the two points A and B of two pipes. The centre of pipe A is 3 m above centre of pipe B. Pipe 'A' contains liquid of specific gravity 1.5, while pipe B contains a liquid of specific gravity 0.9. The manometric liquid mercury is 5m below the centre of pipe A. The pressure at A and B are 1 kgf/cm ² and 1.8 kgf/cm ² respectively. Find the difference in mercury level in the differential manometer.	10	L4	CO1
OR					
Q.2	a.	Derive an expression for total pressure and centre of pressure for a vertical plane surface submerged in liquid.	08	L2	CO2
	b.	What is the bulk modulus of elasticity of a liquid which is compressed in a cylinder from a volume of 0.0125 m ³ at 80 N/cm ² pressure to a volume of 0.0124 m ³ at 150 N/cm ² pressure?	06	L3	CO2
	c.	An equilateral triangular plate of 5m side length is immersed in water with its base and apex at 2 m and 6 m below the free surface of water respectively. Calculate the total force and position of centre of pressure.	06	L4	CO2
Module – 2					
Q.3	a.	Distinguish between : (i) Steady and unsteady flow (ii) Uniform and non-uniform flow (iii) Laminar and turbulent flow	06	L1	CO2
	b.	Derive an expression for continuity equation for a three dimensional flow in Cartesian coordinate.	08	L2	CO2
	c.	In a 2D incompressible flow, the fluid velocity components are given by $u = x - 4y$ and $v = -y - 4x$. Show that velocity potential exists.	06	L3	CO2
OR					
Q.4	a.	State the assumptions and derive Bernoulli's equation of energy along a streamline.	10	L2	CO2
	b.	The following are the data given for laying water supply pipeline. The change in diameter is gradual from 20 cm at 'A' to 50 cm at B. Pressure at A and B is 80 kN/m ² and 60 kN/m ² respectively. The end B is 3m higher than A. If the flow in the pipe is 200 LPS, find: (i) Direction of flow (ii) Head loss between A and B.	10	L4	CO2
Module – 3					
Q.5	a.	Derive an expression for the discharge over a triangular notch.	08	L2	CO3
	b.	Distinguish between pipes in series and pipes in parallel.	04	L1	CO3
	c.	A 0.5 m diameter and 100 m long pipeline carrying 0.5 m ³ /sec of water is fitted with valve at the downstream end. Calculate the rise of pressure caused within the pipe due to valve closure. If: (i) Instantaneously (ii) In one second. Assume sonic velocity as 1430 m/s.	08	L4	CO3

OR

Q.6	a.	Derive Darcy-Weisback equation for head loss due to friction with assumptions.	08	L2	CO3
	b.	Water flows over a rectangular weir 1 m wide at a depth of 150 mm and afterwards passes through a triangular right angled weir. Take C_d for rectangular weir as 0.62 and for triangular weir as 0.59. Find the depth over triangular weir.	08	L3	CO3
	c.	Explain Water Hammer phenomenon.	04	L1	CO3

Module – 4

Q.7	a.	With neat sketches, differentiate between flow through pipes and flow through open channels with examples.	06	L2	CO4
	b.	What is meant by economical section of a channel? Derive the condition for the most economical rectangular section.	08	L1	CO4
	c.	A discharge of $18 \text{ m}^3/\text{sec}$ flows through a rectangular channel 6m wide at a depth of 1.6 m. Find: (i) Specific energy (ii) Critical depth (iii) State whether the flow is subcritical or supercritical	06	L4	CO4

OR

Q.8	a.	Explain the term hydraulic jump. Derive an expression for the depth of hydraulic jump.	10	L2	CO4
	b.	A sluice gate discharges water into a horizontal rectangular channel with a velocity of 6 m/sec and depth of flow is 0.4 m. The width of the channel is 8m. Determine whether a hydraulic jump will occur or not, if occur find its height and loss of energy per kg of water. Also determine the power lost in the hydraulic jump.	10	L4	CO4

Module – 5

Q.9	a.	Explain impulse momentum principle.	02	L2	CO5
	b.	Explain concept of velocity triangles. Also obtain an expression for work done per second by jet striking unsymmetrical moving vane tangentially at one end of the tips.	08	L3	CO5
	c.	Design a pelton wheel turbine required to develop shaft power of 95.6475 KW working under a head of 60 m at a speed of 200 rpm. The overall efficiency may be taken as 85%. Take $C_v = 0.98$ and velocity of the buckets = 0.45 times the velocity of the jet.	10	L4	CO5

OR

Q.10	a.	Draw a neat sketch of the hydro electric power plant. Mention the functions of each component.	08	L2	CO5
	b.	A centrifugal pump is to discharge $0.118 \text{ m}^3/\text{sec}$ at a speed of 1450 rpm against a head of 25 m. The impeller diameter is 250 mm, its width at outlet is 50 mm and manometric efficiency is 75%. Determine the vane angle at the outer periphery of the impeller.	08	L4	CO5
	c.	Distinguish between turbine and pump.	04	L1	CO5

* * * * *

BCV403					
OR					
Q.6	a.	Explain O and D studies objectives and methods of data collections.	12	L3	CO3
	b.	Explain the steps involved in the IRC method of signal design.	8	L2	CO3
Module – 4					
Q.7	a.	With neat sketch, explain permanent way and its requirements.	10	L1	CO4
	b.	Estimate the quantities of materials required to construct 1 km long B.G. track with sleeper density.	10	L3	CO4
OR					
Q.8	a.	Briefly explain the functions and requirements of sleeper and ballast.	10	L1	CO4
	b.	Define station yards. Explain different types of station yards.	10	L2	CO4
Module – 5					
Q.9	a.	Explain the characteristics of an aircraft which affects the planning and design of air ports.	10	L2	CO4
	b.	Write the comparison between runway of highway.	10	L1	CO4
OR					
Q.10	a.	Draw a neat sketch of layout of an airport and explain the functions of various components ports.	10	L2	CO4
	b.	The length of runway under standard conditions is 1700 m. The airport site is at an elevation of 260 m. Its reference temperature 32°C. If the runway is to constructed with an effective gradient of 0.2%. Determine corrected runway length.	10	L3	CO4

CBCS SCHEME

BCV456C

USN

--	--	--	--	--	--	--	--	--	--

Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan. 2025 Electronic Waste Management - Issues and Challenges

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. E – waste is also known as
a) Electronic waste b) Essential waste c) European waste d) Economic waste
 2. E – waste is mainly composed of
a) Valuable metals b) Glass and plastic c) Toxic substances d) all of these
 3. International electronic waste day is held on
a) March 14 b) November 14 c) October 14 d) January 14
 4. What is expanded form of WEEE?
a) Wastes from Electronic and Electrical Equipment
b) Wastes from Environment and Engineering Equipment
c) Wiring Boards of Electrical and Electronic Equipment
d) Wastes from Engineering and Eco-friendly Equipment
 5. China along with _____ produces one – third of the World's e - waste
a) Canada b) USA c) Australia d) Germany
 6. Which international event used medals made with gold extracted from e-waste?
a) ICC Cricket World Cup
b) FIFA World Cup 2022
c) Tokyo Olympics 2020
d) Expo 2020

7. Which one of these is NOT a category of e-waste?
 - a) Screens and monitors
 - b) Plastic bottles
 - c) Heat exchange equipments
 - d) Mobile phones
8. E – Waste constitutes of valuable metals like
 - a) Gold
 - b) Silver
 - c) Copper
 - d) All of these
9. Which of the following elements make e – waste hazardous in nature?
 - a) Plastic
 - b) Iron
 - c) Lead
 - d) Glass
10. Which among the following is a good way to dispose of e-waste?
 - a) Burn it
 - b) Give it to an e-waste - recycler
 - c) Throw it in the street
 - d) none of these
11. What is the iron and steel percentage of e-waste?
 - a) 20%
 - b) 30%
 - c) 40%
 - d) 50%
12. What is the hazardous pollutant released from LED's?
 - a) Arsenic
 - b) Barium
 - c) Cobalt
 - d) Cadmium
13. Identify the hazardous pollutant released from circuit boards?
 - a) Arsenic
 - b) Barium
 - c) Lead
 - d) Copper
14. What happens to old electronic items in poor countries?
 - a) E-waste litters town and leaves a lot of health risks
 - b) E-waste is collected and recycled
 - c) Nothing
 - d) Well managed
15. Which electronic device contributes the most to global e-waste generation?
 - a) Refrigerators
 - b) Washing machines
 - c) Televisions
 - d) Mobile phones
16. What is the primary goal of e-waste management?
 - a) To reduce landfill usage
 - b) To recover valuable materials
 - c) To minimize environmental impacts
 - d) All of these
17. Where does plastic go when we throw it in the dust bin?
 - a) Landfill
 - b) Ocean
 - c) Recycling factory
 - d) All of these
18. Poly Vinyl chloride (PVC) is used in electronic and electrical devices mainly as
 - a) Insulators
 - b) Solders
 - c) TV screens
 - d) Thermostats
19. The unsound practices observed in e-waste sites include
 - a) Scavenging
 - b) Open burning
 - c) Manual disassembly
 - d) All of these
20. Which toxic compound is not found in e-waste?
 - a) Mercury
 - b) Lead
 - c) Neon
 - d) Cadmium

34. Which continent has the highest rate of collection and recycling of e-waste?
a) Asia b) Africa c) Europe d) Australia
35. _____ are obsolete devices from the past that have become outdated (like vacuum tubes) and are new part of e-waste stream.
a) Special e-waste b) Historical e-waste
c) Hazardous waste d) None of these
36. Which factor contributes significantly to the increase in e-waste generation?
a) Rapid technological advancement
b) Longer life span
c) Decrease in device usage
d) Increasing recycling rates
37. What does the '3R' principle in e-waste management stand for?
a) Reclaim, recycle, reduce
b) Reduce, Reuse, Recycle
c) Recovery, Reuse, Reclaim
d) Refurbish, Resell, Rest
38. The first step in e-waste management is
a) Incineration of e-waste
b) Dispose it in landfills
c) Export it to other countries
d) Collect e-waste and segregate it
39. Consider the following statement on e-waste
1) Electronic products that reached its useful life are classified as e-waste
2) E-waste is not harmful to environment if it is properly recycled
a) 1 and 2 are correct
b) Only 1 is correct
c) Only 2 is correct
d) Neither 1 nor 2 are correct
40. Why is e-waste management important?
a) To increase the production of electronic devices
b) To reduce the amount of e-waste generated
c) To encourage improper disposal of e-waste
d) To export e-waste to other countries
41. Which legislation regulates e-waste management in India?
a) The Water Act, 1974
b) Air Act, 1981
c) E-waste (management) Rules, 2022
d) None of these
42. Which year did India implement its first e-waste management rules?
a) 2001 b) 2011 c) 2016 d) 2018
43. Under the E-waste (management) rules, who is responsible for setting up e-waste collection mechanisms?
a) Municipalities b) State Governments
c) Producers of electronic goods d) Consumers

44. Which international convention addresses the transboundary movement of hazardous wastes, including e-wastes?
a) Basel convention
b) Stockholm convention
c) Kyoto protocol
d) Montreal protocol
45. The e-waste management rules, 2022 shall apply to every
a) Manufacturer b) Producer c) Recycler d) All of these
46. What is the penalty framework for non-compliance under the e-waste (management) rules, 2022?
a) A fine of upto Rs. 1000
b) Imprisonment of upto 2 years
c) A Fine of upto Rs. 1 lakh
d) Community service
47. Which of these is a compliance requirement for producers under the E-waste (management) rules, 2016?
a) Setting up collection centers for e-waste
b) Providing buy – back arrangement for old products
c) Both (a) and (b)
d) None of these
48. What additional responsibility do manufacturers and producers have under the 2022 rules?
a) Reporting quarterly returns
b) Developing of better business models
c) Increase their production
d) Ensuring proper disposal of e-waste
49. Despite regulations, e-waste is still illegally transported across borders. Which counties are particularly affected?
a) Developed countries
b) Developing countries
c) Poor and under-developed countries
d) None
50. e-waste (Management) Rules, 2022 came into effect from
a) 1st June 2022
b) 1st December 2022
c) 1st April 2023
d) 1st January 2023

* * * * *

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

BBOK407

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Biology for Engineers

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define cell. Explain the structure and function of plant cell with neat diagram.	08	L2	CO1
	b.	Define Stem Cell. Discuss the types and application of stem cells.	06	L2	CO1
	c.	Describe the properties and functions of hormones.	06	L2	CO1
OR					
Q.2	a.	Discuss the properties and functions of nucleic acids in cellular processes.	07	L2	CO1
	b.	Discuss the properties and functions of enzymes.	07	L2	CO1
	c.	Discuss the properties of vitamins and its supplies.	06	L2	CO1
Module – 2					
Q.3	a.	Apply the knowledge of nucleic acid in DNA finger printing in forensic applications.	08	L3	CO1
	b.	Discuss whey protein and plant based protein as protein based food.	06	L2	CO1
	c.	Write a note on PLA as bioplastic.	06	L1	CO1
OR					
Q.4	a.	Apply your knowledge of lipids and outline the process of obtaining biodiesel from lipids.	07	L3	CO1
	b.	Define vaccine. Discuss the mechanism of RNA vaccine for COVID-19.	07	L2	CO1
	c.	Write a note on enzyme based biosensors.	06	L1	CO1
Module – 3					
Q.5	a.	Compare human brain with computer's CPU.	07	L3	CO2
	b.	Explain lungs as a purification system.	07	L2	CO2
	c.	Write a note on dialysis systems of kidney.	06	L1	CO2
OR					
1 of 2					

Q.6	a.	Illustrate the engineering solutions available for Parkinson's disease.	07	L3	CO2
	b.	Explain heart as a pumping system.	07	L2	CO2
	c.	Write a note on optical correction and materials used for lens.	06	L1	CO2
Module – 4					
Q.7	a.	Illustrate the HBOCs and PFCs as human substituents.	07	L3	CO3
	b.	Explain how the structure of shark skin reduces drag and how these properties have been applied to improve swim suit.	07	L2	CO3
	c.	Explain the term GPS and aircrafts technology as bio inspired by bird fly.	06	L2	CO3
OR					
Q.8	a.	Compare the uses of ultrasonography and sonars.	07	L3	CO3
	b.	Discuss the king fisher beak shaped bullet train to the reduction of noise and improve the stability.	06	L2	CO3
	c.	Explain the term superhydrophobic and self-cleaning in lotus leaf effect.	07	L2	CO3
Module – 5					
Q.9	a.	Explain bioimaging and artificial intelligence technique in disease diagnosis.	08	L2	CO4
	b.	Explain the working principles of electrical tongue and electrical nose in food industry.	06	L2	CO4
	c.	Write a note on bioengineering of Muscular dystrophy and osteoporosis.	06	L1	CO4
OR					
Q.10	a.	Explain the process of biomining via microbial surface adsorption.	07	L2	CO4
	b.	Describe the concept of DNA origami and its role in bio-computing.	06	L2	CO4
	c.	Write a note on self healing bio-concrete and bio-mineralization processes.	07	L1	CO4

CBCS SCHEME

BUHK408

USN

--	--	--	--	--	--	--	--	--	--

Question Paper Version : A

Fourth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Universal Human Values

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

1. What is the state of liking and a holistic and all-encompassing state of the mind that crates inner harmony?
a) Prosperity b) Happiness c) Innateness d) Organized
2. _____ is defined as a group of people working together to create a surplus
a) Alone b) Organisation c) System d) Army
3. The process whereby national economies and business systems are becoming deeply interlinked with each other is called
a) Nationalization b) Glocalization c) Global linking d) Globalization
4. Which of the following actions will likely lead to organizational ethical behavior?
a) Establishing an ethics office
b) Promoting moral courage
c) Developing strong governance processes
d) All of these
5. _____ behavior tends to arise when managers decide to put the attainment of their own personal goals, or the goals of the organization, above the fundamental rights of one or more stakeholder groups
a) Complementary b) Situational c) Unethical d) Confusing
6. Accepted principles of right or wrong governing the conduct of business people are called :
Choose one answer
a) Business b) Conduct c) Business ethics d) Principles
7. What is true about a professional
a) The professional has to have autonomy
b) He cannot be supervised by the client
c) He should not subject to political control
d) All of these

8. The basic rule of professionalism is “not knowingly to do _____”
a) Harm b) Work c) Competition d) Copying
9. The study of _____, as contrasted with legal, obligations is called ethics
a) Behaviors b) Moral c) Business d) All of these
10. Ethics is a conception of _____ conduct
a) Bad/Wrong b) Good/Correct c) Right/True d) Right and Wrong
11. Ethics and _____ go together
a) Profits b) Management c) Worker ship d) All of these
12. _____ driven companies are always successful in the long run
a) Management b) Worker union c) Value d) Profit
13. A business needs to be ethical, in order to
a) Fulfill public expectations for business
b) Prevent harming others
c) Improve business relations and employees productivity
d) All of these
14. Following is (are) types(s) of ethics issues(s)
a) Functional area ethics b) Corporate policy ethics
c) Face to face ethics d) All of these
15. What is called living with assumption for oneself as body and living of human being only on the basis of physical facilities, and not with right understanding and relationship?
a) Human consciousness b) Happiness
c) Right Understanding d) Animal Consciousness
16. Samridhi means
a) Happiness b) Wealth c) Prosperity d) Health
17. Developed nations are the live example of
a) Prosperity b) Wealth c) Happiness d) Health
18. The participation of human beings is seen in two forms
a) Prosperity and work b) Values and Understanding
c) Behavior and Wealth d) Behavior and Work
19. What are the outcomes of realization and understanding?
a) Work b) Values c) Happiness d) Health
20. Corporate culture _____ refers to how widely and deeply employees hold the company’s dominant values and assumptions
a) Breadth b) Depth c) Character d) Strength
21. The observable symbols and signs of a organization’s culture are called :
a) Its cultural design b) Artifacts c) Cultural formatting d) Its cultural dcor
22. _____ is the feeling or responsibility and commitment for ensuring Right Understanding and Right Feeling in the self of my relative
a) Guidance b) Care c) Respect d) Affection
23. The ability to perform a specific task, action, or function successfully is called _____
a) Competence b) Purpose c) Intention d) Program

24. To live in harmony with other human beings is _____.
a) Svatva b) Swatantrata c) Swarajya d) None of these
25. Natural acceptance is?
a) Universal b) Definite c) Continuous d) All of these
26. Basic human aspirations are _____.
a) Happiness b) Prosperity c) Education d) Both (a) and (b)
27. Which of the following is true?
a) Happiness is definite and it can be understood
b) I need to be unhappy to recognize that I am happy
c) One will be bored of happiness if one is always happy
d) I will grow only if I am unhappy. If I become happy my growth will stop
28. To be wealthy is _____ condition in the modern world to be happy
a) Not essential b) True c) Fake d) The most important
29. It is important to have a good _____ to be successful in your career
a) Appear b) Attend c) Attitude d) Respectful
30. _____ skills such as writing, speaking and body language are important work ethics so that you can be clear in your messages to others
a) Communication b) Cooperation c) Honesty d) Integrity
31. You are a person with a high level of _____ if you stand by you stand by your values even if others put pressure on you to do something that is wrong or if they make fun of you
a) Communication b) Cooperation c) Honesty d) Integrity
32. If you don't have _____ you can really get in trouble for lying, cheating or deceiving others
a) Communication b) Cooperation c) Honesty d) Integrity
33. If you have _____ you will be a trusted person because they will see that you are committed to you company
a) Loyalty b) Organizational Skills c) Productivity d) Respect
34. Some jobs cannot be done alone and some can be done much better and faster if you possess the quality of _____.
a) Responsibility b) Respect c) Teamwork d) Cooperation
35. The three major types of ethical issues include except?
a) Communication issues b) Systematic issues
c) Corporate issues d) Individual issues
36. Ethics are moral principles and values which _____.
a) Guide a firm's behavior
b) Govern the actions of an individual
c) Provide employees with rule on how to behave
d) Are legally enforceable
37. The word "Ethics" comes from
a) Ethos b) Eternity c) Elementary d) Essentiality
38. A _____ is a set of principle that people hold themselves to or that are held by organizations or groups
a) Code of ethics b) Code of honor c) Ethical perspective d) Persuasive tactic

39. _____ are beliefs and attitudes we have that can actually conflict with our ethical decision
a) Morals b) Principles c) Virtues d) Values
40. Intellectual property rights are an important aspect in international trade. Which of the following is covered under IPR?
a) Patents b) Copyrights c) Trademarks d) All of these
41. Which of the following is not the infringement of copyright?
a) Copy the software to another computer by educational institution
b) Copy the software to another computer by company
c) Copy the software to another computer
d) To make backup copies
42. Patent can be granted for _____?
a) Process b) Ideas
c) Both product and process d) Product
43. Government may order the non advertisement of any patent application in the case of invention related to _____
a) Information technology b) Food
c) Drug d) Defense technology
44. Patent for the _____ is prohibited
a) Traditional knowledge b) Arms
c) Atomic energy d) Junk food
45. A design can be also registered under _____
a) Trademark Act b) Patent Act
c) Geographical Indication Act d) Copyright Act
46. IPR Complete specification gives _____
a) Full description of claim only
b) Full description of royalty
c) Full description of invention
d) Full description of application
47. Is the first owner of a Book's copyright?
a) Author b) Publisher c) Translator d) Printing press owner
48. Copyright is a _____
a) Creation by person of efforts, intellectual and capital
b) Creation by person of the efforts only
c) Creation by negligence of a person
d) Creation by the mental status of a person
49. Globalization has improved the living structure of which of the following?
a) All the people
b) People living in developing countries
c) People living in developed countries
d) None of these
50. Which sector experienced the least profit due to the effect of Globalization?
a) Industrial sector b) Service sector c) Agriculture Sector d) Secondary sector

* * * * *

--	--	--	--	--	--	--	--	--	--

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Construction Management and Entrepreneurship

Time: 3 hrs.

Max. Marks: 100

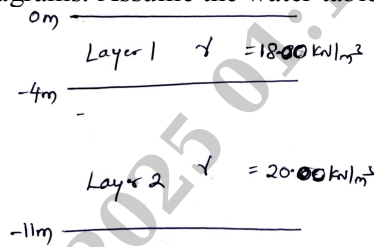
*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1					M	L	C																									
Q.1	a.	Explain in detail construction project formulation.			10	L2	CO1																									
	b.	The activity data of a project is given below: <table border="1"><thead><tr><th>Activity</th><th>Preceding activity</th><th>Duration (Days)</th></tr></thead><tbody><tr><td>A</td><td>-</td><td>05</td></tr><tr><td>B</td><td>-</td><td>15</td></tr><tr><td>C</td><td>-</td><td>09</td></tr><tr><td>D</td><td>A</td><td>06</td></tr><tr><td>E</td><td>C</td><td>04</td></tr></tbody></table> Draw the network diagram, identify the critical path, project duration and free float.			Activity	Preceding activity	Duration (Days)	A	-	05	B	-	15	C	-	09	D	A	06	E	C	04	10	L3	CO1							
Activity	Preceding activity	Duration (Days)																														
A	-	05																														
B	-	15																														
C	-	09																														
D	A	06																														
E	C	04																														
OR																																
Q.2	a.	What is Work Breakdown Structure (WBS)? Mention its significance in construction project.			10	L2	CO1																									
	b.	Below given table pertains to the list of activities and their time estimates of a job: <table border="1"><thead><tr><th>Activity</th><th>Event</th><th>Optimistic time (days)</th><th>Most likely time (days)</th><th>Pessimistic time (days)</th></tr></thead><tbody><tr><td>A</td><td>1 – 2</td><td>3</td><td>7</td><td>10</td></tr><tr><td>B</td><td>1 – 3</td><td>4</td><td>8</td><td>13</td></tr><tr><td>C</td><td>2 – 4</td><td>2</td><td>2</td><td>07</td></tr><tr><td>D</td><td>3 – 4</td><td>5</td><td>8</td><td>10</td></tr></tbody></table> Draw the network and critical path. What is the expected completion time with the probability of 85%? (Take probability factor Z = 1.038)			Activity	Event	Optimistic time (days)	Most likely time (days)	Pessimistic time (days)	A	1 – 2	3	7	10	B	1 – 3	4	8	13	C	2 – 4	2	2	07	D	3 – 4	5	8	10	10	L3	CO1
Activity	Event	Optimistic time (days)	Most likely time (days)	Pessimistic time (days)																												
A	1 – 2	3	7	10																												
B	1 – 3	4	8	13																												
C	2 – 4	2	2	07																												
D	3 – 4	5	8	10																												
Module – 2																																
Q.3	a.	Discuss on Class of Labour. What are the key factors of minimum wages act 1948?			10	L2	CO2																									
	b.	List the factors affecting Labour productivity? Briefly discuss any three factors.			10	L2	CO2																									
OR																																
Q.4	a.	Enumerate the factors to be considered for selection of Construction Equipment.			10	L2	CO2																									
	b.	Explain material management and inventory management.			10	L2	CO2																									
Module – 3																																
Q.5	a.	Explain types of procurement and procurement planning.			10	L2	CO3																									
	b.	Explain the sustainable procurement management.			10	L2	CO3																									
OR																																
Q.6	a.	Explain the different types of construction contracts.			10	L2	CO3																									
	b.	Define contractor and subcontractor. Explain the effective sub contractor management.			10	L2	CO3																									

Module – 4					
Q.7	a.	Explain the process of construction project quality management.	10	L2	CO4
	b.	Explain the safety measures adopted during construction.	10	L2	CO4
OR					
Q.8	a.	Explain Safety Management and Risk Management.	10	L2	CO4
	b.	Explain the terms: i) Facilities Management ii) Occupancy certificate	10	L2	CO4
Module – 5					
Q.9	a.	Explain the different characteristics of a Successful Entrepreneur.	10	L2	CO5
	b.	Explain 5M model and communication skills.	10	L2	CO5
OR					
Q.10	a.	Explain the Business Planning process, Marketing Plan and Financial Plan.	10	L2	CO5
	b.	Explain the role and significance of venture capital.	10	L2	CO5

* * * * *

OR

Q.4	a.	What is meant by total stress, neutral stress and effective stress?	6	L1	CO2
	b.	What is flow net? Mention its applications.	6	L2	CO2
	c.	For the soil deposit shown below, draw the total stress, pore water pressure and effective stress diagrams. Assume the water table is at ground level.  Fig.Q.4(c)	8	L3	CO2

Module – 3

Q.5	a.	Differentiate between standard proctor test and modified proctor test.	6	L2	CO2												
	b.	What are the effects of compaction on soil properties? Explain.	6	L1	CO2												
	c.	The following data was obtained from a proctor compaction test: <table border="1" data-bbox="379 929 1200 1003"> <tr> <td>Water content (%)</td><td>5.90</td><td>7.50</td><td>9.7</td><td>11.65</td><td>13.85</td></tr> <tr> <td>Weight of wet sample (N)</td><td>18.20</td><td>19.50</td><td>20.10</td><td>20.00</td><td>19.70</td></tr> </table> Assume $G = 2.7$, volume of mould = $9.5 \times 10^{-4} \text{ m}^3$. Plot the compaction curve. Determine OMC and MDD. Also plot Zero Air voids line.	Water content (%)	5.90	7.50	9.7	11.65	13.85	Weight of wet sample (N)	18.20	19.50	20.10	20.00	19.70	8	L3	CO2
Water content (%)	5.90	7.50	9.7	11.65	13.85												
Weight of wet sample (N)	18.20	19.50	20.10	20.00	19.70												

OR

Q.6	a.	Explain the concept of consolidation by mass spring analogy.	6	L2	CO2
	b.	Explain the Terzaghi's consolidation theory with its limitations.	6	L2	CO2
	c.	The time to reach 40% consolidation of a two way drained laboratory 12 mm thick saturated clayey soil sample is 40 sec. Determine the time required for 60% consolidation of the same soil of 10 m thick on the top of a rocky surface subjected to the same loading conditions as laboratory sample.	8	L3	CO2

Module – 4

Q.7	a.	Explain assumptions of Mohr's strength theory and mention its limitations.	6	L2	CO3
	b.	Explain the factors affecting shear strength of soil.	6	L2	CO3
	c.	An unconfined compression test was conducted on an undisturbed sample of clay. The sample had a diameter of 38 mm and was 80 mm long. The load at failure measured as 30 N and the axial deformation of the sample at failure was 12 mm. Determine the unconfined compressive strength and undrained shear strength of clay, if failure angle = 50° .	8	L3	CO3

OR

Q.8	a.	Briefly explain different drainage conditions of triaxial shear test.	6	L2	CO3															
	b.	Mention any 4 advantages of triaxial shear test.	4	L1	CO3															
	c.	A consolidated undrained test was carried out on a clayey sample and the results are as follows. Find total and effective shear parameters of soil. <table><tr><td>Cell pressure, kN/m²</td><td>100</td><td>200</td><td>400</td><td>600</td></tr><tr><td>Deviator stress, kN/m²</td><td>300</td><td>410</td><td>610</td><td>850</td></tr><tr><td>Pore water pressure, kN/m³</td><td>-45</td><td>-15</td><td>50</td><td>110</td></tr></table>	Cell pressure, kN/m ²	100	200	400	600	Deviator stress, kN/m ²	300	410	610	850	Pore water pressure, kN/m ³	-45	-15	50	110	10	L3	CO3
Cell pressure, kN/m ²	100	200	400	600																
Deviator stress, kN/m ²	300	410	610	850																
Pore water pressure, kN/m ³	-45	-15	50	110																

Module – 5

Q.9	a.	Explain the assumptions of Terzaghi's bearing capacity theory.	6	L2	CO4
	b.	Differentiate between general shear failure, local shear failure, punching shear failure.	6	L2	CO4
	c.	What will be the gross and net safe bearing pressure of sand having $\phi = 40^\circ$, unit weight of sand = 19 kN/m ³ under i) 1.2 m wide strip footing ii) 1.2 m square footing. Assume the footings are placed at a depth of 1.2 m below G.L and water table is at greater depth. Also assume F.O.S. = 3 and $N_c = 95.7$, $N_q = 81.3$, $N_y = 100.4$.	8	L3	CO4

OR

Q.10	a.	Differentiate immediate settlement, consolidation and secondary settlements.	6	L2	CO4
	b.	Give the tolerance limits as per BIS specifications for total and differential settlements for footing and rafts.	8	L2	CO4
	c.	A clayey stratum of 5 m thick has a unit weight of 15 kN/m ³ water content of 43% liquid limit = 80%, specific gravity $G = 2.7$. Initial overburden pressure due to old structure is 300 kN/m ² , due to construction of a building stress increased to 120 kN/m ² . Determine the consolidation settlement.	6	L3	CO4

* * * * *

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

BCV503

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Concrete Technology

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. Use IS : 456-2000, IS 10262 : 2019 are permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the chemical composition of ordinary Portland cement.	10	L2	CO1
	b.	Describe : i) Rapid Hardening Cement (RHC) ii) Sulphate Resisting Cement (SRC)	10	L2	CO1
OR					
Q.2	a.	Describe briefly the classification of Aggregates.	10	L2	CO1
	b.	Explain the effects of fly ash and silica fresh concrete.	10	L2	CO1
Module – 2					
Q.3	a.	Briefly explain which are the factors affecting workability.	10	L2	CO2
	b.	Briefly, explain any 2 lab test to measure the workability of fresh concrete.	10	L2	CO2
OR					
Q.4	a.	Briefly explain the necessity of compaction of concrete. Also explain different methods such as Hand Compaction and compaction by needle vibrator.	10	L2	CO2
	b.	Explain the need for curing of concrete. Also what is water curing and membrane curing?	10	L2	CO2
Module – 3					
Q.5	a.	Explain creep of concrete and factors affecting creep of concrete.	10	L2	CO3
	b.	Explain Shrinkage of concrete. Also explain plastic shrinkage and drying shrinkage of concrete.	10	L2	CO3
OR					
Q.6	a.	Describe Sulphate attack and Chloride attack on Hardened concrete.	10	L2	CO3
	b.	Explain : i) Rebound Hammer test ii) Ultrasonic pulse velocity.	10	L2	CO3
Module – 4					
Q.7		Explain Significance of concrete mix design and write the steps involved in concrete mix design as per IS code and also discuss the variables in proportioning of concrete.	20	L2	CO4

OR					
Q.8		Design a concrete mix for grade M ₂₅ a) Grade designation → M-25 b) Type of cement → OPC 53 grade c) Maximum nominal Aggregate size → 20mm d) Minimum cement content → 310 Kg/m ³ e) Maximum water cement ratio → 0.45 f) Workability → 50-75 mm (Slump) g) Exposure condition → Normal h) Degree of supervision → Good i) Type of aggregate → Crushed angular aggregate j) Maximum cement content → 540 Kg/m ³ k) Chemical admixture type → Super plasticizer l) Specific gravity of cement → 3.15 m) Specific gravity of water → 1.0 n) Specific gravity of C.A → 2.882 o) Water absorption of C.A → 1% p) Free surface moisture : Nil q) Specific gravity of fine aggregate : 2.605 r) Water absorption of fine aggregate : 1.23% s) Free surface moisture of F.A : Nil.	20	L3	CO4
Module – 5					
Q.9	a.	Explain the manufacturing process of Ready mix concrete.	10	L2	CO5
	b.	Explain the concept of Self Compacting Concrete (SCC). And its advantages and disadvantages of SCC.	10	L2	CO5
OR					
Q.10	a.	What is light weight concrete? Explain different materials used in light weight concrete.	10	L2	CO5
	b.	Explain : i) High strength concrete ii) High performance concrete.	10	L2	CO5

* * * * *

CBCS SCHEME

USN

--	--	--	--	--	--	--	--	--	--

BCV515C

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Solid Waste Management

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Briefly, discuss the policies and Legislative framework of solid waste management.	10	L2	CO1
	b.	Briefly explain the classification in solid waste management.	10	L2	CO1
OR					
Q.2	a.	Discuss the Role of stakeholders in solid waste management.	10	L2	CO2
	b.	Discuss the need of solid waste management and explain the concept of 3R.	10	L2	CO2
Module – 2					
Q.3	a.	Explain the various factors affecting the solid waste generation.	10	L2	CO2
	b.	Briefly, explain the physical properties of municipal solid waste.	10	L2	CO2
OR					
Q.4	a.	Briefly, explain the different methods to estimate the quantity of waste generated.	10	L2	CO2
	b.	Briefly, explain the chemical and Biological properties of solid waste.	10	L2	CO2
Module – 3					
Q.5	a.	Distinguish between Haulid container system and stationary container system with a schematic diagram.	10	L2	CO2
	b.	What are transfer stations? Briefly explain any two types of Transfer station.	10	L1	CO2
OR					
Q.6	a.	Briefly, discuss the types of solid waste collection services and outline the importance of onsite processing.	10	L2	CO2
	b.	Briefly explain the following : i) Collection vehicle ii) Types of storage containers.	10	L2	CO2
Module – 4					
Q.7	a.	Define RDF. Explain the steps unvalued in RDF.	10	L1	CO3
	b.	What is composting? Explain the Bangalore method of computing with neat sketch.	10	L1	CO3

OR					
Q.8	a.	What are the factors to be considered while designing a land fill?	10	L2	CO3
	b.	What is incineration? Discuss the advantages and disadvantages of incineration?	10	L2	CO5
Module – 5					
Q.9	a.	Briefly explain the effects and disposal methods of E-waste.	10	L2	CO3
	b.	Explain the effects and disposal of construction and demolition waste.	10	L1	CO3
OR					
Q.10	a.	Explain the classification of Biomedical wastes.	10	L1	CO3
	b.	Explain the Life Cycle Assessment in waste management.	10	L1	CO3

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

Question Paper Version : A

Fifth Semester B.E./B.Tech Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Which of the following is a Producer?
 a) Animals b) Human beings c) Plants d) Fish
 2. An ecosystem consists of
 a) Biotic species b) Biotic species and physical factors
 c) Only insects d) Only materials
 3. The term Ecology was coined by
 a) E.P. Odium b) A. G. Tansley c) Albert Einstein d) A.G. Haeckal
 4. Which year did the concept of sustainability first appear?
 a) 1992 b) 1978 c) 1980 d) 1987
 5. According to United Nations, the following are the significant issues of social progress.
 a) Instruction b) Public Health c) Living standards d) All of these
 6. The number of SDGS are
 a) 10 b) 15 c) 17 d) 16
 7. Which one of the following is not comes under SDG's?
 a) Zero Hunger b) Political activity c) Education d) Gender quality
 8. Which of the following has the largest population in food chain?
 a) Producers b) 1st degree consumers
 c) 2nd degree consumers d) Decomposers
 9. U.N era of scheduling for sustainable growth is from
 a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14

10. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake
11. Solar Energy is an example of
a) Renewable b) Continuous c) Non - renewable d) None of these
12. Hydrogen energy can be tapped through
a) Heat pumps b) Fuel cells c) Petroleum d) Gassifiers
13. OTEC is an energy technology that converts
a) Energy in large tides of ocean to generate electricity
b) Energy in ocean waves to generate electricity
c) Energy in ocean due to thermal gradient to generate electricity
d) Energy in the fast moving ocean currents to generate electricity.
14. Which of the following source of energy is less eco – friendly?
a) Biogas b) Wind c) Solar d) Nuclear
15. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
16. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
17. Identify the renewable sourced of energy from the following :
a) Coal b) Wind power c) Uranium d) Oil
18. Which of the following is biotic?
a) Forest b) Iron c) Water d) None
19. World Environment Day celebrated on
a) 6th June b) 5th July c) 5th June d) 5th Jan
20. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these
21. Temporary hardness of water is due to
a) Chloride hardness b) Manganese hardness
c) Calcium hardness d) Carbonate hardness
22. The green house gas is
a) N₂O b) CH₄ c) CO₂ d) All of these
23. Which is not a natural source of pollution?
a) Volcanoes b) Forest - Fire c) Coal - fire d) Dust storms
24. Air pollution from automobiles can be controlled by providing
a) Wet collector b) Scrubbers c) Catalytic converter d) All of these
25. Normal sound level of quiet speech is
a) 120 dB b) 90 dB c) 140 dB d) 50 dB

26. What is the permissible range of p^H for drinking water as per the Indian standard?
 a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8.5 d) 6.5 to 8.5
27. Which of the following is a major cause of soil pollution?
 a) Accidents involving the vehicles that are transporting waste material.
 b) Pesticides and chemical fertilizers from agricultural lands
 c) Improper solid waste disposal d) All of these
28. What does E – Waste stand for?
 a) Environment Waste b) Electronic Waste
 c) Equipment Waste d) All of these
29. What is municipal solid waste disposal?
 a) The placement on the land b) The lack of waste
 c) In sanitary land fill site d) All of these
30. What is major cause of air pollution?
 a) Fossil fuel burning b) Industry emission
 c) Wild fire d) All of these
31. Average long – term weather of an area is called
 a) Weather conditions b) Seasonal variations
 c) Average weather d) Climate
32. Climate and global air circulations are mainly affected by the properties of
 a) Water and Air b) Temperature c) Precipitation d) None of these
33. The p^H of acid rain is
 a) less than 5.2 b) 5.2 – 6.2 c) 6.2 – 7.2 d) Above 7.2
34. Acid rain has been increasing day by day due to
 a) Urbanisation b) Industrialisation
 c) Increase in vehicle population d) None of these
35. Which is responsible for Ozone depletion?
 a) Methyl bromide b) CFC's
 c) Hydro chlorofluorocarbons d) All of these
36. The International protocol to protect the Ozone layer's
 a) Kytoto protocol b) Basal protocol c) Montreal protocol d) Viema protocol
37. The Ozone layer is locate upto _____ km above the earth surface.
 a) 1000 b) 50 c) 80 d) 100
38. Acid rain effects on
 a) Materials b) Plants c) Soil d) All of these
39. High – Concentration of fluoride content in drinking water causes.
 a) Dental flurosis b) Bone brittling c) Dental caries d) Plague
40. Desired concentration of fluoride content in drinking water in mg/ℓ is
 a) 0.7 – 1.2 b) 0.5 – 2 c) 2 – 5 d) 1 – 1.75

41. The First United National conference as Human Environment was held at Stickholm in
a) December 1972 b) June 1972 c) June 1974 d) June 1992
42. To achieve the goal of clean environment, important strategies required are
a) Effective laws b) Active participation of the publics
c) Active participation of NGO's d) Both (a) and (b)
43. There are provisions for protection for our environment under
a) Indian Penal Code b) Police Act c) Municipal Act d) All of these
44. The Government of India enacted the Water Act in the year.
a) 1970 b) 1974 c) 1975 d) 1980
45. The Environmental Act of India was enacted in the year.
a) 1986 b) 1992 c) 1984 d) 1974
46. The Environmental Act, 1986 deals with
a) Air b) Water c) Land d) All of these
47. Karnataka State "Pollution Control Board" was established in the year.
a) 1974 b) 1982 c) 1986 d) 1976
48. Which of the following is not a waste treatment method for biomedical waste?
a) Incineration b) Chemical disinfecting
c) Auto claving d) Sieving
49. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation
50. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

Question Paper Version : B

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. The First United National conference as Human Environment was held at Stickholm in
a) December 1972 b) June 1972 c) June 1974 d) June 1992
 2. To achieve the goal of clean environment, important strategies required are
a) Effective laws b) Active participation of the publics
c) Active participation of NGO's d) Both (a) and (b)
 3. There are provisions for protection for our environment under
a) Indian Penal Code b) Police Act c) Municipal Act d) All of these
 4. The Government of India enacted the Water Act in the year.
a) 1970 b) 1974 c) 1975 d) 1980
 5. The Environmental Act of India was enacted in the year.
a) 1986 b) 1992 c) 1984 d) 1974
 6. The Environmental Act, 1986 deals with
a) Air b) Water c) Land d) All of these
 7. Karnataka State "Pollution Control Board" was established in the year.
a) 1974 b) 1982 c) 1986 d) 1976
 8. Which of the following is not a waste treatment method for biomedical waste?
a) Incineration b) Chemical disinfecting
c) Auto claving d) Sieving
 9. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation

10. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron
11. Average long – term weather of an area is called
a) Weather conditions b) Seasonal variations
c) Average weather d) Climate
12. Climate and global air circulations are mainly affected by the properties of
a) Water and Air b) Temperature c) Precipitation d) None of these
13. The p^H of acid rain is
a) less than 5.2 b) 5.2 – 6.2 c) 6.2 – 7.2 d) Above 7.2
14. Acid rain has been increasing day by day due to
a) Urbanisation b) Industrialisation
c) Increase in vehicle population d) None of these
15. Which is responsible for Ozone depletion?
a) Methyl bromide b) CFC's
c) Hydro chlorofluorocarbons d) All of these
16. The International protocol to protect the Ozone layer's
a) Kytoto protocol b) Basal protocol c) Montreal protocol d) Viema protocol
17. The Ozone layer is locate upto _____ km above the earth surface.
a) 1000 b) 50 c) 80 d) 100
18. Acid rain effects on
a) Materials b) Plants c) Soil d) All of these
19. High – Concentration of fluoride content in drinking water causes.
a) Dental flurosis b) Bone brittling c) Dental caries d) Plague
20. Desired concentration of fluoride content in drinking water in mg/ℓ is
a) 0.7 – 1.2 b) 0.5 – 2 c) 2 – 5 d) 1 – 1.75
21. Solar Energy is an example of
a) Renewable b) Continuous c) Non - renewable d) None of these
22. Hydrogen energy can be tapped through
a) Heat pumps b) Fuel cells c) Petroleum d) Gassifiers
23. OTEC is an energy technology that converts
a) Energy in large tides of ocean to generate electricity
b) Energy in ocean waves to generate electricity
c) Energy in ocean due to thermal gradient to generate electricity
d) Energy in the fast moving ocean currents to generate electricity.
24. Which of the following source of energy is less eco – friendly?
a) Biogas b) Wind c) Solar d) Nuclear

25. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
26. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
27. Identify the renewable sourced of energy from the following :
a) Coal b) Wind power c) Uranium d) Oil
28. Which of the following is biotic?
a) Forest b) Iron c) Water d) None
29. World Environment Day celebrated on
a) 6th June b) 5th July c) 5th June d) 5th Jan
30. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these
31. Which of the following is a Producer?
a) Animals b) Human beings c) Plants d) Fish
32. An ecosystem consists of
a) Biotic species b) Biotic species and physical factors
c) Only insects d) Only materials
33. The term Ecology was coined by
a) E.P. Odiyum b) A. G. Tansley c) Albert Einstein d) A.G. Haeckal
34. Which year did the concept of sustainability first appear?
a) 1992 b) 1978 c) 1980 d) 1987
35. According to United Nations, the following are the significant issues of social progress.
a) Instruction b) Public Health c) Living standards d) All of these
36. The number of SDGS are
a) 10 b) 15 c) 17 d) 16
37. Which one of the following is not comes under SDG's?
a) Zero Hunger b) Political activity c) Education d) Gender quality
38. Which of the following has the largest population in food chain?
a) Producers b) 1st degree consumers
c) 2nd degree consumers d) Decomposers
39. U.N era of scheduling for sustainable growth is from
a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14
40. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

Question Paper Version : C

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Temporary hardness of water is due to
 - a) Chloride hardness
 - b) Manganese hardness
 - c) Calcium hardness
 - d) Carbonate hardness
 2. The green house gas is
 - a) N_2O
 - b) CH_4
 - c) CO_2
 - d) All of these
 3. Which is not a natural source of pollution?
 - a) Volcanoes
 - b) Forest - Fire
 - c) Coal - fire
 - d) Dust storms
 4. Air pollution from automobiles can be controlled by providing
 - a) Wet collector
 - b) Scrubbers
 - c) Catalytic converter
 - d) All of these
 5. Normal sound level of quiet speech is
 - a) 120 dB
 - b) 90 dB
 - c) 140 dB
 - d) 50 dB
 6. What is the permissible range of p^H for drinking water as per the Indian standard?
 - a) 6 to 9
 - b) 6.5 to 7.5
 - c) 6 to 8.5
 - d) 6.5 to 8.5
 7. Which of the following is a major cause of soil pollution?
 - a) Accidents involving the vehicles that are transporting waste material.
 - b) Pesticides and chemical fertilizers from agricultural lands
 - c) Improper solid waste disposal
 - d) All of these
 8. What does E – Waste stand for?
 - a) Environment Waste
 - b) Electronic Waste
 - c) Equipment Waste
 - d) All of these

9. What is municipal solid waste disposal?
a) The placement on the load b) The lack of waste
c) In sanitary land fill site d) All of these
10. What is major cause of air pollution?
a) Fossil fuel burning b) Industry emission
c) Wild fire d) All of these
11. The First United National conference as Human Environment was held at Stockholm in
a) December 1972 b) June 1972 c) June 1974 d) June 1992
12. To achieve the goal of clean environment, important strategies required are
a) Effective laws b) Active participation of the publics
c) Active participation of NGO's d) Both (a) and (b)
13. There are provisions for protection for our environment under
a) Indian Penal Code b) Police Act c) Municipal Act d) All of these
14. The Government of India enacted the Water Act in the year.
a) 1970 b) 1974 c) 1975 d) 1980
15. The Environmental Act of India was enacted in the year.
a) 1986 b) 1992 c) 1984 d) 1974
16. The Environmental Act, 1986 deals with
a) Air b) Water c) Land d) All of these
17. Karnataka State "Pollution Control Board" was established in the year.
a) 1974 b) 1982 c) 1986 d) 1976
18. Which of the following is not a waste treatment method for biomedical waste?
a) Incineration b) Chemical disinfecting
c) Auto claving d) Sieving
19. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation
20. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron
21. Average long – term weather of an area is called
a) Weather conditions b) Seasonal variations
c) Average weather d) Climate
22. Climate and global air circulations are mainly affected by the properties of
a) Water and Air b) Temperature c) Precipitation d) None of these
23. The p^H of acid rain is
a) less than 5.2 b) 5.2 – 6.2 c) 6.2 – 7.2 d) Above 7.2

24. Acid rain has been increasing day by day due to
 - a) Urbanisation
 - b) Industrialisation
 - c) Increase in vehicle population
 - d) None of these
25. Which is responsible for Ozone depletion?
 - a) Methyl bromide
 - b) CFC's
 - c) Hydro chlorofluorocarbons
 - d) All of these
26. The International protocol to protect the Ozone layer's
 - a) Kytoto protocol
 - b) Basal protocol
 - c) Montreal protocol
 - d) Viema protocol
27. The Ozone layer is locate upto _____ km above the earth surface.
 - a) 1000
 - b) 50
 - c) 80
 - d) 100
28. Acid rain effects on
 - a) Materials
 - b) Plants
 - c) Soil
 - d) All of these
29. High – Concentration of fluoride content in drinking water causes.
 - a) Dental flurosis
 - b) Bone brittling
 - c) Dental caries
 - d) Plague
30. Desired concentration of fluoride content in drinking water in mg/ℓ is
 - a) 0.7 – 1.2
 - b) 0.5 – 2
 - c) 2 – 5
 - d) 1 – 1.75
31. Solar Energy is an example of
 - a) Renewable
 - b) Continuous
 - c) Non - renewable
 - d) None of these
32. Hydrogen energy can be tapped through
 - a) Heat pumps
 - b) Fuel cells
 - c) Petroleum
 - d) Gassifiers
33. OTEC is an energy technology that converts
 - a) Energy in large tides of ocean to generate electricity
 - b) Energy in ocean waves to generate electricity
 - c) Energy in ocean due to thermal gradient to generate electricity
 - d) Energy in the fast moving ocean currents to generate electricity.
34. Which of the following source of energy is less eco – friendly?
 - a) Biogas
 - b) Wind
 - c) Solar
 - d) Nuclear
35. O₃ Represents
 - a) Oxygen
 - b) Planets
 - c) Ozone
 - d) Green house gases
36. Which of the following is an example of Water Borne Diseases
 - a) Hepatitis - A
 - b) Hepatitis - B
 - c) Hepatitis - C
 - d) Hepatitis - H
37. Identify the renewable soured of energy from the following :
 - a) Coal
 - b) Wind power
 - c) Uranium
 - d) Oil
38. Which of the following is biotic?
 - a) Forest
 - b) Iron
 - c) Water
 - d) None
39. World Environment Day celebrated on
 - a) 6th June
 - b) 5th July
 - c) 5th June
 - d) 5th Jan

40. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these
41. Which of the following is a Producer?
a) Animals b) Human beings c) Plants d) Fish
42. An ecosystem consists of
a) Biotic species b) Biotic species and physical factors
c) Only insects d) Only materials
43. The term Ecology was coined by
a) E.P. Odiyum b) A. G. Tansley c) Albert Einstein d) A.G. Haeckel
44. Which year did the concept of sustainability first appear?
a) 1992 b) 1978 c) 1980 d) 1987
45. According to United Nations, the following are the significant issues of social progress.
a) Instruction b) Public Health c) Living standards d) All of these
46. The number of SDGS are
a) 10 b) 15 c) 17 d) 16
47. Which one of the following is not comes under SDG's?
a) Zero Hunger b) Political activity c) Education d) Gender quality
48. Which of the following has the largest population in food chain?
a) Producers b) 1st degree consumers
c) 2nd degree consumers d) Decomposers
49. U.N era of scheduling for sustainable growth is from
a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14
50. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake

* * * * *

USN

--	--	--	--	--	--	--	--	--	--

Question Paper Version : D

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025 Environmental Studies

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Average long – term weather of an area is called
 - a) Weather conditions
 - b) Seasonal variations
 - c) Average weather
 - d) Climate
 2. Climate and global air circulations are mainly affected by the properties of
 - a) Water and Air
 - b) Temperature
 - c) Precipitation
 - d) None of these
 3. The p^H of acid rain is
 - a) less than 5.2
 - b) 5.2 – 6.2
 - c) 6.2 – 7.2
 - d) Above 7.2
 4. Acid rain has been increasing day by day due to
 - a) Urbanisation
 - b) Industrialisation
 - c) Increase in vehicle population
 - d) None of these
 5. Which is responsible for Ozone depletion?
 - a) Methyl bromide
 - b) CFC's
 - c) Hydro chlorofluorocarbons
 - d) All of these
 6. The International protocol to protect the Ozone layer's
 - a) Kytoto protocol
 - b) Basal protocol
 - c) Montreal protocol
 - d) Viema protocol
 7. The Ozone layer is locate upto _____ km above the earth surface.
 - a) 1000
 - b) 50
 - c) 80
 - d) 100
 8. Acid rain effects on
 - a) Materials
 - b) Plants
 - c) Soil
 - d) All of these
 9. High – Concentration of fluoride content in drinking water causes.
 - a) Dental flurosis
 - b) Bone brittling
 - c) Dental caries
 - d) Plague

10. Desired concentration of fluoride content in drinking water in mg/l is
a) 0.7 – 1.2 b) 0.5 – 2 c) 2 – 5 d) 1 – 1.75
11. Which of the following is a Producer?
a) Animals b) Human beings c) Plants d) Fish
12. An ecosystem consists of
a) Biotic species b) Biotic species and physical factors
c) Only insects d) Only materials
13. The term Ecology was coined by
a) E.P. Odium b) A. G. Tansley c) Albert Einstein d) A.G. Haeckel
14. Which year did the concept of sustainability first appear?
a) 1992 b) 1978 c) 1980 d) 1987
15. According to United Nations, the following are the significant issues of social progress.
a) Instruction b) Public Health c) Living standards d) All of these
16. The number of SDGS are
a) 10 b) 15 c) 17 d) 16
17. Which one of the following is not comes under SDG's?
a) Zero Hunger b) Political activity c) Education d) Gender quality
18. Which of the following has the largest population in food chain?
a) Producers b) 1st degree consumers
c) 2nd degree consumers d) Decomposers
19. U.N era of scheduling for sustainable growth is from
a) 2002 - 11 b) 2003 - 12 c) 2004 - 13 d) 2005 - 14
20. Which of the following is an example of Artificial Ecosystem?
a) Forest b) River c) Aquarium d) Lake
21. The First United National conference as Human Environment was held at Stockholm in
a) December 1972 b) June 1972 c) June 1974 d) June 1992
22. To achieve the goal of clean environment, important strategies required are
a) Effective laws b) Active participation of the publics
c) Active participation of NGO's d) Both (a) and (b)
23. There are provisions for protection for our environment under
a) Indian Penal Code b) Police Act c) Municipal Act d) All of these
24. The Government of India enacted the Water Act in the year.
a) 1970 b) 1974 c) 1975 d) 1980
25. The Environmental Act of India was enacted in the year.
a) 1986 b) 1992 c) 1984 d) 1974
26. The Environmental Act, 1986 deals with
a) Air b) Water c) Land d) All of these

27. Karnataka State "Pollution Control Board" was established in the year.
a) 1974 b) 1982 c) 1986 d) 1976
28. Which of the following is not a waste treatment method for biomedical waste?
a) Incineration b) Chemical disinfecting
c) Auto claving d) Sieving
29. The Organic material of the solid waste will decompose.
a) By the flow of water b) By the soil particles
c) By the action of micro organisms d) B Oxidation
30. Which of the following element make e – waste hazardous in nature?
a) Lead b) Glass c) Plastic d) Iron
31. Temporary hardness of water is due to
a) Chloride hardness b) Manganese hardness
c) Calcium hardness d) Carbonate hardness
32. The green house gas is
a) N_2O b) CH_4 c) CO_2 d) All of these
33. Which is not a natural source of pollution?
a) Volcanoes b) Forest - Fire c) Coal - fire d) Dust storms
34. Air pollution from automobiles can be controlled by providing
a) Wet collector b) Scrubbers c) Catalytic converter d) All of these
35. Normal sound level of quiet speech is
a) 120 dB b) 90 dB c) 140 dB d) 50 dB
36. What is the permissible range of p^H for drinking water as per the Indian standard?
a) 6 to 9 b) 6.5 to 7.5 c) 6 to 8.5 d) 6.5 to 8.5
37. Which of the following is a major cause of soil pollution?
a) Accidents involving the vehicles that are transporting waste material.
b) Pesticides and chemical fertilizers from agricultural lands
c) Improper solid waste disposal d) All of these
38. What does E – Waste stand for?
a) Environment Waste b) Electronic Waste
c) Equipment Waste d) All of these
39. What is municipal solid waste disposal?
a) The placement on the land b) The lack of waste
c) In sanitary land fill site d) All of these
40. What is major cause of air pollution?
a) Fossil fuel burning b) Industry emission
c) Wild fire d) All of these
41. Solar Energy is an example of
a) Renewable b) Continuous c) Non - renewable d) None of these

42. Hydrogen energy can be tapped through
a) Heat pumps b) Fuel cells c) Petroleum d) Gassifiers
43. OTEC is an energy technology that converts
a) Energy in large tides of ocean to generate electricity
b) Energy in ocean waves to generate electricity
c) Energy in ocean due to thermal gradient to generate electricity
d) Energy in the fast moving ocean currents to generate electricity.
44. Which of the following source of energy is less eco – friendly?
a) Biogas b) Wind c) Solar d) Nuclear
45. O₃ Represents
a) Oxygen b) Planets c) Ozone d) Green house gases
46. Which of the following is an example of Water Borne Diseases
a) Hepatitis - A b) Hepatitis - B c) Hepatitis - C d) Hepatitis - H
47. Identify the renewable sourced of energy from the following :
a) Coal b) Wind power c) Uranium d) Oil
48. Which of the following is biotic?
a) Forest b) Iron c) Water d) None
49. World Environment Day celebrated on
a) 6th June b) 5th July c) 5th June d) 5th Jan
50. Which of the following is conventional source of Energy?
a) Wind Energy b) Solar Energy c) Bio gas d) None of these

* * * * *

--	--	--	--	--	--	--	--	--	--

Fifth Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025
Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
 2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Identify the meaning of Research and brief out the objective and motivation in engineering research.	10	L1	CO1
	b.	Explain brief about research cycle and verify with the research flow diagram.	10	L1	CO1
OR					
Q.2	a.	Identify the types of engineering research and briefly explain them.	10	L1	CO1
	b.	Explain about the different types of research misconduct.	10	L1	CO1
Module – 2					
Q.3	a.	Explain about the importance of literature review and technical reading.	10	L2	CO2
	b.	Mention the various benefits of bibliographic databases.	10	L1	CO2
OR					
Q.4	a.	Identify the impact of technical reaction and brief about it.	10	L1	CO2
	b.	Enumerate the impact of title and keywords on citation with example.	10	L2	CO2
Module – 3					
Q.5	a.	Define Intellectual properties and explain about its types.	10	L1	CO3
	b.	Explain about the key aspect of patent law.	10	L2	CO3
OR					
Q.6	a.	Explain about the assessment of novelty.	10	L1	CO3
	b.	Brief about the patent procedure in India.	10	L1	CO4
Module – 4					
Q.7	a.	Mention and brief about the justification for copyright law.	10	L2	CO4
	b.	Explain about the basic concepts of under lying copyright law.	10	L1	CO4
OR					
Q.8	a.	Brief about the various representations of sound recordings.	10	L2	CO5
	b.	Explain about TRIPS agreement in detail.	10	L1	CO5

Module – 5					
Q.9	a.	Explain about the justification of protection designs.	10	L2	CO5
	b.	Brief about the excluded subjected matter in the context of design protection.	10	L1	CO5
OR					
Q.10	a.	What are the rights of the owner of designs? Explain.	10	L1	CO5
	b.	Brief about the Assignment of Design Rights.	10	L1	CO5
